

labadmin / homelab

# Home Lab Technical Walkthrough

How the three-node Proxmox lab, its services and its safety gates fit together

|             |                          |
|-------------|--------------------------|
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Written for an engineer who has never seen this lab. Every file mention is a link, every code excerpt is pulled from the repository at build time, and the build fails if an excerpt no longer matches its source.

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# 1 Overview

This lab is three Proxmox VE 9 nodes in one cluster called `homelab`, a Synology NAS that holds the backups and the private DNS zone, and a Windows desktop (workstation) used only for on-demand workloads. The repository you are reading is the only way changes reach it: a gated, stage-by-stage toolkit of Ansible, OpenTofu and Python. Nothing runs end to end. Every live stage is one Make target that refuses to start until the operator passes an approval flag and an exact confirmation string.

## NOTE

Facts in this document come from `docs/reference/environment.md` (not published), `MEMORY.md` (not published) and the source itself, and are true as of the commit printed on the cover. Volatile state (what is running today, open gaps) lives in `MEMORY.md` (not published) and `HOMELAB-ROADMAP.md` (not published); read those before acting.

## 1.1 Hosts

| Host        | Address      | Alias       | Role                                                            | Hardware                                     |
|-------------|--------------|-------------|-----------------------------------------------------------------|----------------------------------------------|
| pve1        | 192.168.0.11 | pve1        | Cluster seed; EVE-NG, observability, bao-1                      | Xeon E-2186M, 12 cores, 31 GiB, 477 GiB NVMe |
| pve2        | 192.168.0.12 | pve2        | Homelab services, bao-2                                         | i5-8250U, 8 cores, 11.5 GiB, 932 GiB HDD     |
| pve3        | 192.168.0.13 | pve3        | NetBox, bao-3                                                   | i5-6500, 4 cores, 7.6 GiB, 112 GiB SSD       |
| nas1        | 192.168.0.20 | nas         | Synology the NAS: backups (NFS), split-horizon DNS, media share | ARM64, 1.6 GiB                               |
| workstation | 192.168.0.70 | workstation | Windows 11 Home: Plex, media guest, music library, staging      | Ryzen 7 2700X, 32 GiB, GTX 1080              |
| PiVPN       | 192.168.0.60 | pivpn       | WireGuard, IPsec, Pi-hole (unmanaged)                           | Raspberry Pi                                 |

The network is one flat `/22` (`192.168.0.0/24`) behind an the router router. The `192.168.0.x` addresses are inside that same `/22`; they are **not** a separate security zone. The single source of truth for every identity above is `inventory/lab.yml`:

```
14 schema_version: 2
15
16 network:
17   cidr: 192.168.0.0/24           # one flat /22; 192.168.0.x is NOT a separate zone
18   gateway: 192.168.0.1         # the router; never authoritative for the lab zone
19   bridge: vmbr0                # not VLAN-aware on any node
20   dns_zone: lab.example.com
21   lab_resolver: 192.168.0.20    # Synology split-horizon DNS
22   ntp: pool.ntp.org
```

Source: `inventory/lab.yml`, lines 14–22

The network block is what every consumer (OpenTofu through `yamlddecode()` , the Ansible inventories through a contract test) agrees on. The comments carry the two constraints that matter most: the /22 is flat, and the the router must never become authoritative for the lab zone. A `null` anywhere in this file means “not recorded”, and any consumer that needs the value refuses to run.

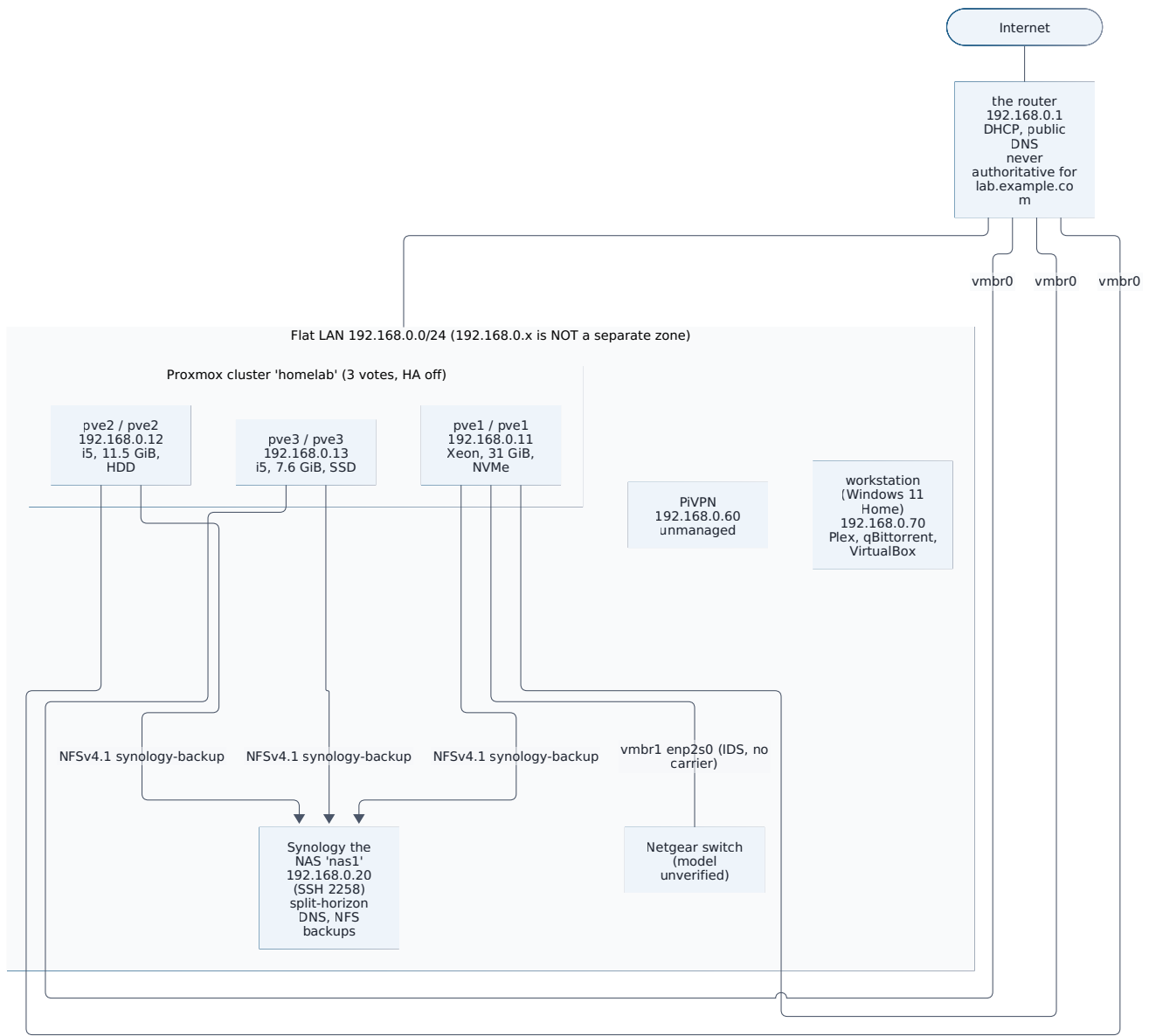


Figure 1. Physical hosts, addresses and the flat /22 network

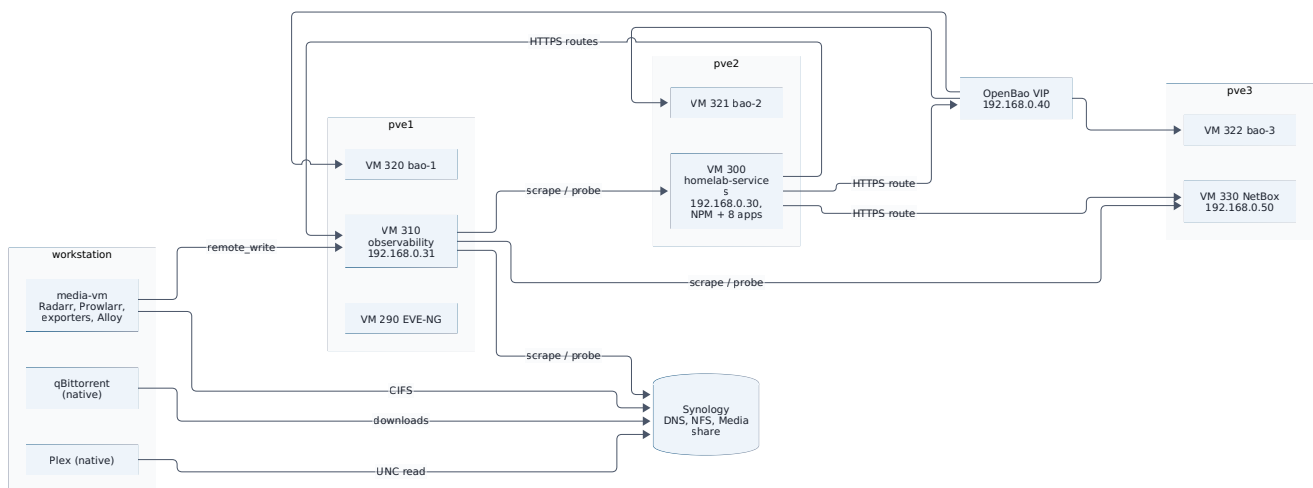
Diagram source: [docs/walkthrough/diagrams/01-physical-network.mmd](#)

## 1.2 Service catalogue

| Guest  | VMID | Node | Address | What runs there                                             | Component                         |
|--------|------|------|---------|-------------------------------------------------------------|-----------------------------------|
| EVE-NG | 290  | pve1 | —       | Network emulation lab (restored from vzdump, never rebuilt) | <a href="#">platform/proxmox/</a> |

| Guest               | VMID       | Node         | Address                  | What runs there                                                                                                   | Component                                                                                 |
|---------------------|------------|--------------|--------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| homelab-services    | 300        | pve2         | 192.168.0.30             | Nginx Proxy Manager, Homepage, Uptime Kuma, Linkding, IT-Tools, PairDrop, Speedtest Tracker, NetKnife (API + web) | <a href="#">services/homelab-services/</a>                                                |
| observability       | 310        | pve1         | 192.168.0.31             | Prometheus, Grafana, Loki, Alertmanager, Alloy and five exporters                                                 | <a href="#">observability/</a>                                                            |
| bao-1, bao-2, bao-3 | 320-322    | one per node | 192.168.0.41-22, VIP .19 | OpenBao 2.6.1 Raft cluster, <b>un-initialized</b>                                                                 | <a href="#">security/secrets-open-bao/</a>                                                |
| netbox              | 330        | pve3         | 192.168.0.50             | NetBox (six containers)                                                                                           | <a href="#">services/net-box/</a>                                                         |
| media-vm            | VirtualBox | workstation  | NAT                      | Radarr, Prowlarr, Recyclarr, exporters, Alloy                                                                     | <a href="#">services/media/</a>                                                           |
| (native)            | —          | workstation  | —                        | Plex Media Server, qBittorrent, Music Library                                                                     | <a href="#">services/media/</a> , <a href="#">services/music-library/</a> (not published) |

Private sites (`*.lab.example.com`) all terminate TLS at Nginx Proxy Manager on VM 300 with a trusted DNS-01 wildcard certificate, then forward to the guest that serves them.



**Figure 2.** Logical components and how they depend on each other

Diagram source: [docs/walkthrough/diagrams/02-logical-components.mmd](#)

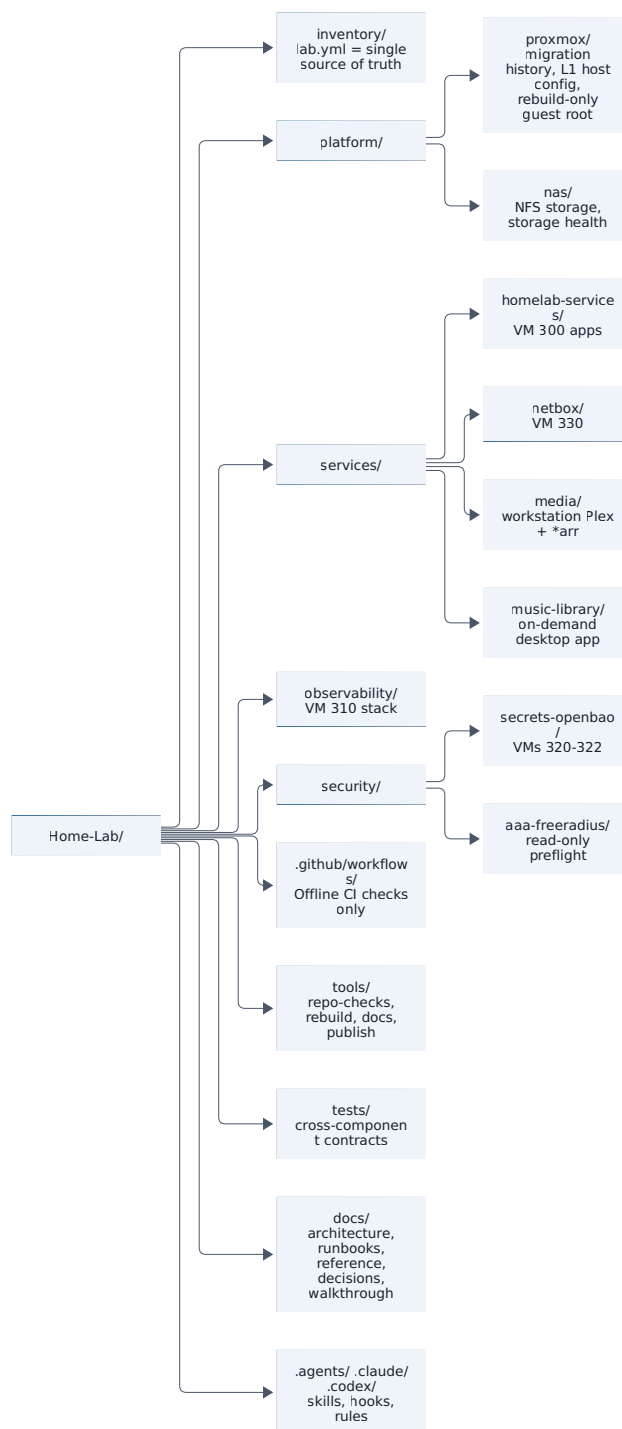
## 2 Repository map

The monorepo groups components by layer. Each component has its own `Makefile`, `AGENTS.md` and offline checks; the root `Makefile` only dispatches one target to one component (`make COMPONENT=<path> <target>`).

|                       |                                                          |
|-----------------------|----------------------------------------------------------|
| └─ inventory/         | lab.yml (single source of truth) + Ansible inventories   |
| └─ platform/          |                                                          |
| └─ proxmox/           | cluster migration history, guest OpenTofu, rebuild L0–L3 |
| └─ nas/               | Synology NFS preflight and registration                  |
| └─ services/          |                                                          |
| └─ homelab-services/  | VM 300 applications                                      |
| └─ netbox/            | VM 330 NetBox                                            |
| └─ media/             | workstation Plex and the *arr guest                      |
| └─ music-library/     | local music library (desktop + web companion)            |
| └─ observability/     | VM 310 stack, dashboards, alerts, backups                |
| └─ security/          |                                                          |
| └─ secrets-openbao/   | OpenBao cluster, PKI, policies, backups                  |
| └─ aaa-freeradius/    | FreeRADIUS discovery (no deployment yet)                 |
| └─ tests/             | cross-component safety contracts                         |
| └─ tools/             | repo checks, rebuild planner, CI helper, docs, publisher |
| └─ docs/              | decisions, runbooks, reference, walkthrough              |
| └─ .github/workflows/ | offline CI, PR plans, dispatch-only deploys              |

| Path                                    | Responsibility                                                  | Invoked by                                                                | Reads / writes                                                  |
|-----------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------|
| <code>inventory/lab.yml</code>          | Every node, guest, address, MAC, backup job                     | OpenTofu <code>yamlldecode()</code> , contract tests, rebuild planner     | Read only by tooling; edited by hand                            |
| <code>inventory/ *.yml</code>           | Ansible inventories per component                               | Each component's <code>ansible.cfg</code>                                 | Read by Ansible                                                 |
| <code>platform/proxmox/</code>          | Migration stages 0–11, VM 300/310 adoption, rebuild L0–L3       | <code>make -C platform/proxmox &lt;stage&gt;</code>                       | Proxmox hosts over SSH; artifacts/ (git-ignored)                |
| <code>platform/nas/</code>              | NFS target preflight and storage registration                   | <code>make -C platform/nas preflight</code>                               | Proxmox storage config                                          |
| <code>services/homelab-services/</code> | VM 300 base, one Compose project per app, app backups           | <code>make -C platform/proxmox homelab-service SERVICE=&lt;app&gt;</code> | <code>/opt/homelab</code> , <code>/srv/homelab</code> on VM 300 |
| <code>services/netbox/</code>           | VM 330 guest, NetBox Compose, <code>pg_dump</code> backups      | <code>make -C platform/proxmox netbox-*</code>                            | VM 330                                                          |
| <code>services/media/</code>            | Media guest, *arr containers and settings, Plex checks          | <code>make -C services/media &lt;stage&gt;</code>                         | media-vm, NAS media share                                       |
| <code>observability/</code>             | Central monitoring stack and its backup/restore drill           | <code>make -C observability &lt;stage&gt;</code>                          | VM 310                                                          |
| <code>security/secrets-openbao/</code>  | OpenBao hosts, TLS, HA, post-init policy, Raft backups          | <code>make -C security/secrets-openbao &lt;stage&gt;</code>               | VMs 320–322                                                     |
| <code>tests/</code>                     | Contracts across components: gates, workflows, denylist, guards | <code>make COMPONENT=tests check, CI</code>                               | Read only                                                       |

| Path                                                                           | Responsibility                                                                   | Invoked by                        | Reads / writes                                         |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------|
| <a href="#">tools/</a>                                                         | Inventory contract, forbidden constructs, rebuild planner, docs build, publisher | Make targets and CI               | Generated docs only                                    |
| <a href="#">.github/workflows/</a> (not published)                             | Offline checks only                                                              | GitHub Actions                    | See <a href="#">CI checks</a>                          |
| <a href="#">.agents/</a> , <a href="#">.claude/</a> , <a href="#">.co-dex/</a> | Agent skills, guard hook, execution policy                                       | Coding agents                     | Read only                                              |
| <a href="#">docs/</a>                                                          | Decisions, runbooks, reference, this walkthrough                                 | Humans, <a href="#">make docs</a> | <a href="#">docs/reference/generated/</a> is generated |



**Figure 3.** Which directory is responsible for which part of the lab

Diagram source: [docs/walkthrough/diagrams/03-repo-responsibility.mmd](#)

## 2.1 One source of truth

`inventory/lab.yml` is the single source of truth (ADR-0005). A live identity changes there first and in its consumer in the same commit. `tools/repo-checks/inventory_contract.py` compares the file with every Ansible inventory and OpenTofu root. Deliberate differences are recorded in a ratchet:

```

21 # Drift that exists on purpose, with the reason and the plan to remove it.
22 # A ratchet: an entry that no longer drifts must be deleted (stale_known_drift).
23 KNOWN_DRIFT = {

```

```
24     (
25         "services/netbox/terraform/proxmox-guest/main.tf",
26         "dns_server",
27     ): "Live root still names the NAS's retired .20 address; changing it would make "
28     "a plan against VM 330 propose a cloud-init change. Fix with the state migration "
29     "(ADR-0003), then delete this entry.",
30 }
```

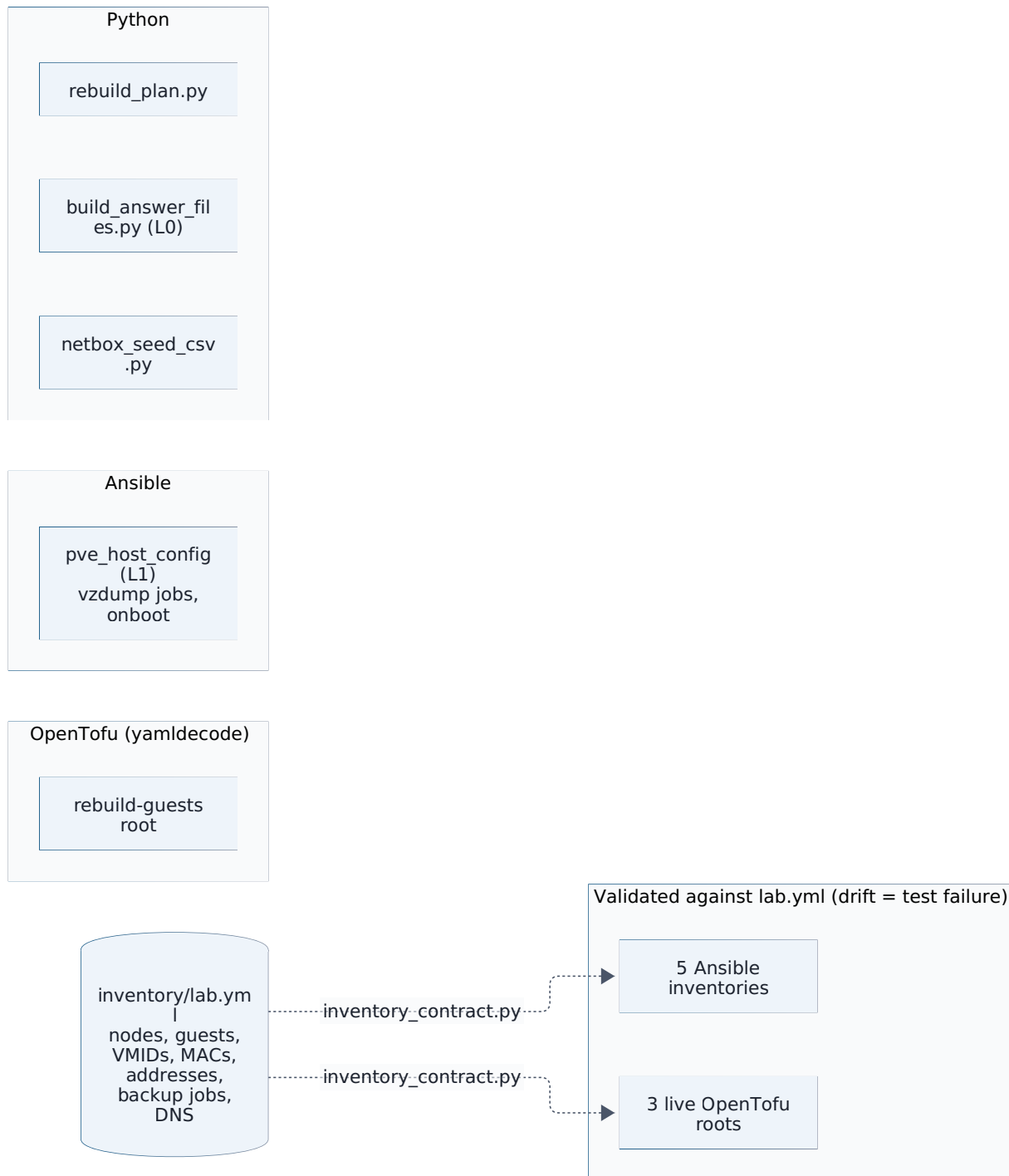
Source: [tools/repo-checks/inventory\\_contract.py](#), lines 21-30

Each entry is keyed by the consumer file and the key that differs, and its value says why the difference is allowed and how it will be removed. The one current entry is the NetBox root's retired NAS address. Changing it before the state migration would make a plan against the live VM 330 propose a cloud-init change.

```
229 def unexpected(drift) -> list[str]:
230     return [f"{w}: {k}: {d}" for w, k, d in drift if (w, k) not in KNOWN_DRIFT]
231
232
233 def stale_known_drift(drift) -> list[str]:
234     present = {(w, k) for w, k, _ in drift}
235     return [
236         f"{w}: {k}: no longer drifts; remove it from KNOWN_DRIFT"
237         for (w, k) in KNOWN_DRIFT
238         if (w, k) not in present
239     ]
```

Source: [tools/repo-checks/inventory\\_contract.py](#), lines 229-239

The ratchet works in both directions. `unexpected` returns drift that is not listed, which fails `test_no_unexpected_drift`. `stale_known_drift` returns entries that no longer drift, which fails `test_known_drift_entries_are_still_real`. An exception therefore cannot outlive its reason.

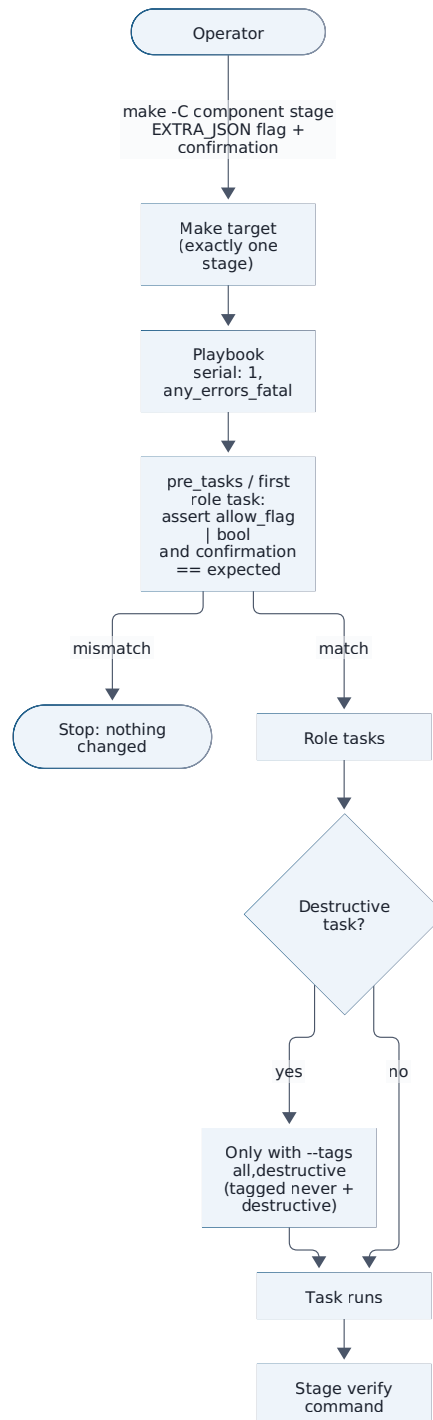


**Figure 4.** inventory/lab.yml and the consumers checked against it

Diagram source: <docs/walkthrough/diagrams/04-single-source-of-truth.mmd>

### 3 Safety model

The lab is live and some of its data is irreplaceable. The repository is built so that the default outcome of any command is “nothing changed”. Every rule below has a mechanism that enforces it and a test that fails the build if the mechanism is removed.



**Figure 5.** How one gated stage runs, from the operator’s command to the verify step

Diagram source: <docs/walkthrough/diagrams/05-gated-stage.mmd>

| Rule                                                                                         | Gate or mechanism                                                                                                          | Enforcing test                                                                                                                                         |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| A live stage needs an approval flag <b>and</b> an exact confirmation                         | <code>pre_tasks assert</code> in each playbook; flags default <code>false</code> , confirmations <code>""</code>           | <code>tests/test_stage_gates.py</code> , <code>test_every_allow_flag_defaults_to_false</code> , <code>test_every_confirmation_defaults_to_empty</code> |
| Destructive tasks never run by accident                                                      | Tagged <code>never + destructive</code> ; need <code>--tags all,destructive</code>                                         | <code>test_destructive_tasks_carry_never_and_destructive</code>                                                                                        |
| One host at a time, stop on first error                                                      | <code>serial: 1</code> , <code>any_errors_fatal: true</code>                                                               | <code>test_every_playbook_is_serial_one_and_fatal_on_error</code>                                                                                      |
| No aggregate target ( <code>deploy-all</code> and friends)                                   | Root Makefile only dispatches one target                                                                                   | <code>test_makefile_has_no_aggregate_target</code> , <code>test_no_target_chains_rungs</code>                                                          |
| Backups are never pruned or deleted                                                          | NFS storage and every job are <code>keep-all=1</code>                                                                      | <code>test_no_archive_is_ever_deleted</code> , <code>test_pruning_is_disabled_by_policy</code> , <code>test_no_backup_job_prunes</code>                |
| OpenTofu never destroys a live guest                                                         | <code>prevent_destroy</code> , <code>purge_on_destroy = false</code> , import-first adoption; no apply/destroy Make target | <code>test_home_lab_terraform_is_import_first_and_destroy_protected</code> , <code>checks-only workflow contract</code>                                |
| SSH host keys are never auto-accepted                                                        | <code>StrictHostKeyChecking</code> stays on in every <code>ansible.cfg</code>                                              | <code>test_every_monorepo_ansible_cfg_keeps_host_key_checking_on</code>                                                                                |
| No secret is ever tracked                                                                    | <code>.gitignore</code> , gitleaks in pre-commit and CI, denylist scan                                                     | <code>test_no_secret_bearing_path_is_tracked</code> , <code>test_no_secret_shaped_value_is_tracked</code>                                              |
| No <code>ignore_errors</code> , <code>rm -rf</code> or <code>StrictHostKeyChecking=no</code> | Forbidden-construct scanner with a shrinking baseline                                                                      | <code>test_tree_has_no_forbidden_construct_beyond_the_baseline</code>                                                                                  |
| Agents cannot run live commands or read secrets                                              | Claude <code>PreToolUse</code> guard, Codex execution-policy rules                                                         | <code>test_forbidden_commands_are_blocked</code> , <code>test_forbidden_commands</code>                                                                |
| GitHub performs hosted checks only                                                           | Read permissions, SHA-pinned actions, no secrets or deploy stages                                                          | <code>checks-only workflow contract</code>                                                                                                             |
| Every stateful item has a backup or a stated reason                                          | Coverage and backup matrices                                                                                               | <code>test_zero_undocumented_rows</code> , <code>test_every_guest_has_a_backup_job_or_a_reason</code>                                                  |

### 3.1 The approval gate

Every live playbook opens the same way. Here is the stage that deploys one application to VM 300:

```

1  ---
2  - name: Deploy exactly one reviewed Home Lab application
3    hosts: homelab_service_guests
4    gather_facts: false
5    become: true
6    serial: 1
7    any_errors_fatal: true
8    # Phase 3: every live stage is gated. Extra vars override these defaults.
9    vars:
10     homelab_allow_service_deploy: false
11     homelab_service_deploy_confirmation: ""
12    pre_tasks:
13     - name: Require this stage's approval and its exact confirmation

```

```

14     ansible.builtin.assert:
15         that:
16             - homelab_allow_service_deploy | bool
17             -
18     homelab_service_deploy_confirmation == ('DEPLOY ' ~ homelab_services_service_name ~ ' ON ' ~
19     inventory_hostname)
20     fail_msg: "Set homelab_allow_service_deploy=true and homelab_service_deploy_confirmation
21     to the exact string in this playbook."
22     quiet: true
23
24 roles:
25     - role: homelab_services

```

Source: [services/homelab-services/ansible/playbooks/40-deploy-service.yml](#), lines 1-22

The play targets one group, one host at a time (`serial: 1`), and aborts every host on the first failure (`any_errors_fatal`). The flag and the confirmation are declared as play variables with safe defaults, so the assert fails unless both are overridden with extra variables. The confirmation embeds the application name and the host, so an approval for one app on one guest cannot be replayed against another. The test below pins this exact shape.

```

51 @pytest.mark.parametrize("playbook", sorted(GATES))
52 def test_stage_is_gated_before_any_role(root, playbook):
53     flag, confirmation, expected = GATES[playbook]
54     (play,) = yaml.safe_load((root / playbook).read_text(encoding="utf-8"))
55     assert play["vars"][flag] is False
56     assert play["vars"][confirmation] == ""
57     first = play["pre_tasks"][0]["ansible.builtin.assert"]["that"]
58     assert first == [f"{flag} | bool", f"{confirmation} == ({expected})"]

```

Source: [tests/test\\_stage\\_gates.py](#), lines 51-58

The test loads the playbook as YAML rather than grepping it. It checks that the defaults are `False` and `""` and that the **first** pre-task is the two-line assert, so no role can run before the gate. Confirmations are passed as `EXTRA_JSON`, never as `-e key=value`: Ansible's key/value parser splits multi-word strings (see [lessons learned](#)).

## 3.2 Destructive tasks

### **DANGER — DESTRUCTIVE STAGE**

Tasks that stop guests, wipe disks or discard data carry both the `never` and `destructive` tags. They run only when the command line also says `--tags all,destructive`, the stage's `allow_*` flag is `true`, and its exact confirmation string matches. Never add the tags to a command you have not reviewed line by line.

```

16 - name: Require the destructive opt-in tag as well as the approval variables
17     ansible.builtin.assert:
18         that:
19             - "'destructive' in ansible_run_tags or 'never' in ansible_run_tags"
20     fail_msg: |
21         The approval variables are set, but the destructive tasks in this play are
22         tagged `never,destructive` and were therefore not requested.
23
24     Re-run the same command with the explicit opt-in:
25     --tags all,destructive
26     (the Makefile targets for destructive stages already pass this).
27     success_msg: "Destructive opt-in present (--tags all,destructive)."

```

Source: [platform/proxmox/ansible/roles/vzdump\\_backup/tasks/shutdown\\_guests.yml](#), lines 16–27

The second assert exists because Ansible silently skips `never`-tagged tasks. Without it, an operator who set the approval variables but forgot the tag would see a green run that shut nothing down and backed up running guests. The assert turns that silent skip into a failure that says exactly what to add.

```

60 - name: Request a graceful ACPI/agent shutdown of every running guest
61   ansible.builtin.command:
62     argv: [qm, shutdown, "{{ item.key }}", --timeout, "{{ guest_shutdown_timeout_seconds }}", --
forceStop, "0"]
63   changed_when: true
64   loop: "{{ vzdump_backup_original_states | dict2items }}"
65   loop_control:
66     label: "{{ item.key }} ({{ item.value }})"
67   when: item.value == 'running'
68   register: vzdump_backup_shutdown_results
69   tags: [never, destructive]

```

Source: [platform/proxmox/ansible/roles/vzdump\\_backup/tasks/shutdown\\_guests.yml](#), lines 60–69

The shutdown itself is graceful (`--forceStop 0`), loops only over guests that were recorded as running, and is tagged so it cannot run by default. The recorded power states are written to `artifacts/` `first`, so the rollback knows which guests to start again.

### 3.3 Structural tests

```

132 def test_every_playbook_is_serial_one_and_fatal_on_error(repo_root):
133     problems = []
134     for path in sorted((repo_root / "ansible/playbooks").rglob("*.yaml")):
135         for play in _plays(path):
136             if not isinstance(play, dict):
137                 continue
138             name = f"{path.name}::{play.get('name', '?')}"
139             if play.get("any_errors_fatal") is not True:
140                 problems.append(f"{name}: any_errors_fatal is not true")
141             targets_hosts = play.get("hosts") not in ("localhost", "127.0.0.1")
142             if targets_hosts and play.get("serial") != 1:
143                 problems.append(f"{name}: serial is not 1")
144         assert not problems, "\n".join(problems)
145
146
147 def test_no_unattended_aggregate_playbook_exists(repo_root):
148     names = {p.name for p in (repo_root / "ansible/playbooks/migration").glob("*.yaml")}
149     for banned in ("site.yaml", "main.yaml", "all.yaml", "40-deploy.yaml", "migrate.yaml"):
150         assert banned not in names
151
152
153 def test_makefile_has_no_aggregate_target(repo_root):
154     text = (repo_root / "Makefile").read_text()
155     for banned in ("deploy-all:", "migrate-all:", "run-all:", "everything:"):
156         assert banned not in text, f"aggregate target {banned} must not exist"

```

Source: [platform/proxmox/tests/test\\_repo\\_safety.py](#), lines 132–156

These tests read every playbook and the Makefile. They fail on a play without `any_errors_fatal: true`, a host play without `serial: 1`, a banned playbook name such as `site.yaml`, or an aggregate target such as `deploy-all:`. Local plays against `localhost` are exempt from `serial` because they touch no host.

### 3.4 Agent guardrails

Coding agents work in this repository. The Claude Code hook [.claude/hooks/guard.py](#) and the Codex rules in [.codex/rules/homelab.rules](#) block live commands and reads of secret-bearing paths:

```

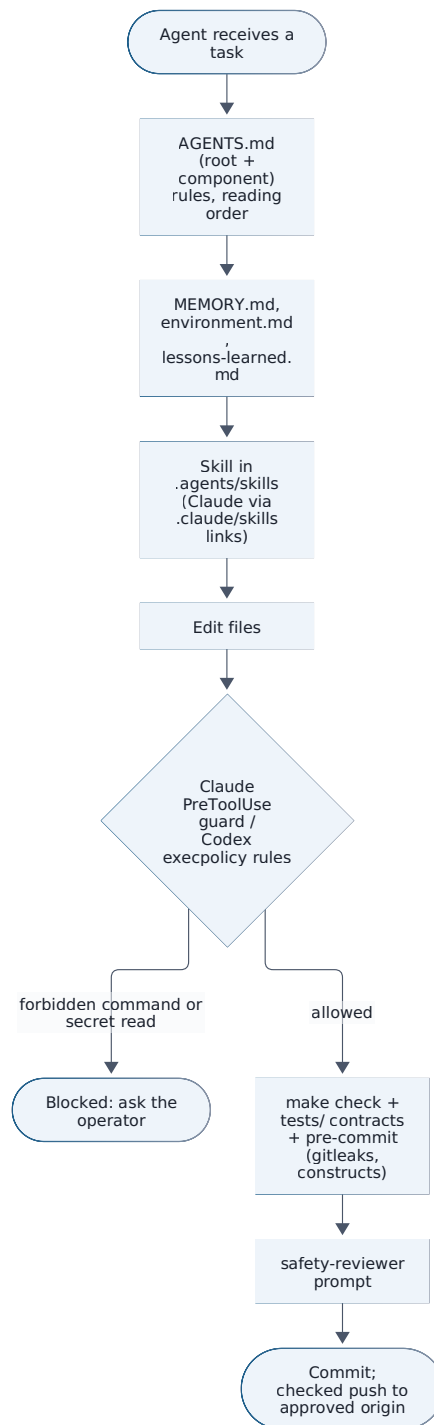
24 PROTECTED_PATH = re.compile(
25     r"""(?ix)
26     (?:\.tfstate(?:\.backup)?$|\.tfstate\.)
27     | (?^\[[/\\])\^[/\\]*\.env$
28     | (?^\[[/\\])\.env\.(!example$)[/\\]+$
29     | \.tfvars(?:\.json)?$
30     | \.(?:pem|key|p12|pfx)$
31     | (?^\[[/\\])id_(?:rsa|ed25519|ecdsa|dsa)$
32     | _ed25519$
33     | (?^\[[/\\~])\.ssh(?:[/\\]|$)
34     | (?^\[[/\\])artifacts(?:[/\\]|$)
35     | (?^\[[/\\])host-configs(?:[/\\]|$)
36     | (?^\[[/\\])Preferences\.xml$
37     | (?^\[[/\\])config\.xml$
38     | (?^\[[/\\])\.local-credentials(?:[/\\]|$)
39     | (?^\[[/\\])secrets(?:[/\\]|$)
40     | (?^\[[/\\])\.runtime[/\\]radarr(?:[/\\]|$)
41     | (?^\[[/\\])auth\.json$
42     """
43 )

```

# OpenTofu/Terraform state  
# real .env files  
# .env.<name>, not .env.example  
# real tfvars  
# private keys and bundles  
# SSH private keys  
# named private keys  
# ~/.ssh/\*\*  
# machine captures  
# host configuration captures  
# Plex token store  
# \*arr API keys  
# operator secret store  
# runtime secret directories  
# quarantined legacy state  
# agent credentials

Source: [.claude/hooks/guard.py](#), lines 24–43

The guard blocks reading these paths; listing a filename is still allowed. It fails closed: malformed hook input, or any internal error, blocks the tool call. [tests/test\\_claude\\_guard.py](#) runs the hook as a subprocess against forbidden commands, wrapped forms (`ssh`, `sudo`, `bash -c`, `xargs`), and ordinary offline work that must stay allowed.



**Figure 6.** How an agent change moves from task to commit under the guardrails

Diagram source: <docs/walkthrough/diagrams/17-agent-workflow.mmd>

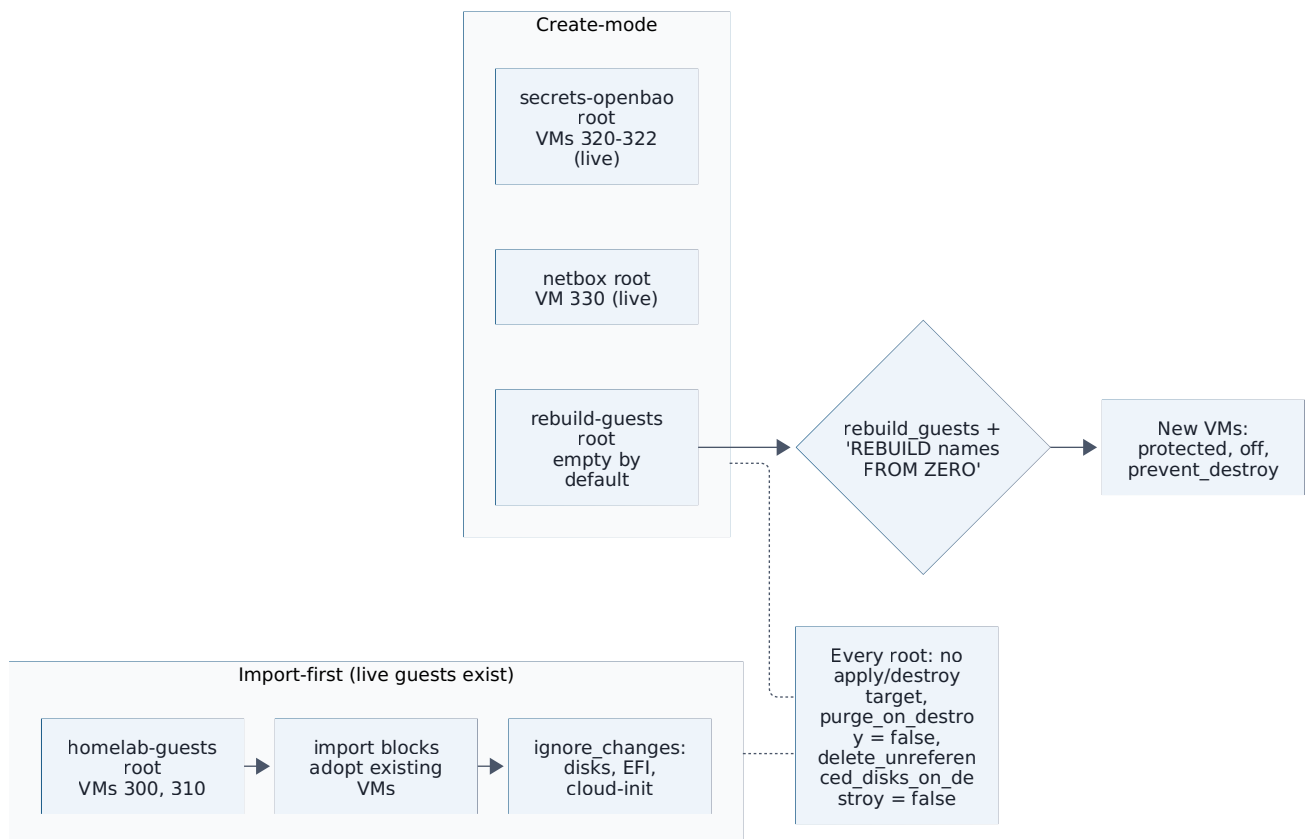
The full list of gates, generated from the playbooks, is in [Appendix C](#).

## 4 Proxmox cluster

### 4.1 Architecture

Three standalone PVE hosts were converted into one cluster, `homelab`, without losing a protected guest (stages 0-11, all complete). HA is disabled. There is no shared storage for guests: `pve1` uses `local` (directory on NVMe), and `pve2` and `pve3` each have their own `local-lvm` thin pool, so live migration is not available. Shared storage exists only for backups: `synology-backup`, an NFS export on NAS Volume 1 registered with `--prune-backups keep-all=1`.

OpenTofu owns guest identity (VMID, node, CPU, memory, NIC, protection, startup). It does not own disks, EFI, cloud-init, applications, DNS or passwords. Two kinds of root exist ([ADR-0007](#)): import-first roots that adopt live guests, and one create-mode root that is only ever used to rebuild after a total loss.



**Figure 7.** Import-first roots adopt live guests; the rebuild-only root creates them after a loss

Diagram source: `docs/walkthrough/diagrams/07-tofu-import-vs-create.mmd`

### 4.2 Files

| File                                                       | Purpose                                                              |
|------------------------------------------------------------|----------------------------------------------------------------------|
| <code>platform/proxmox/Makefile</code>                     | Every stage as one target; <code>EXTRA_JSON</code> carries approvals |
| <code>platform/proxmox/ansible/playbooks/migration/</code> | Numbered migration stages 00-90                                      |
| <code>platform/proxmox/ansible/roles/</code>               | Backup, verify, cluster, join, host-config and guest roles           |

| File                                                       | Purpose                                                                      |
|------------------------------------------------------------|------------------------------------------------------------------------------|
| <a href="#">platform/proxmox/terraform/homelab-guests/</a> | Import-first adoption of VMs 300 and 310                                     |
| <a href="#">platform/proxmox/terraform/rebuild-guests/</a> | Create-mode root, driven by <code>lab.yml</code>                             |
| <a href="#">platform/proxmox/scripts/</a>                  | Answer files, restore plans, capacity, VMID collisions, archive verification |
| <a href="#">platform/proxmox/tests/test_repo_safety.py</a> | The structural safety suite                                                  |
| <a href="#">platform/proxmox/docs/</a>                     | Migration runbook, rollback, troubleshooting, rebuild catalogue              |

## 4.3 Key code

```

1 # Import-first adoption prevents an empty Terraform state from proposing new
2 # VMs over the two live, protected guests.
3 import {
4   to = proxmox_virtual_environment_vm.homelab_services
5   id = "pve2/300"
6 }
7
8 import {
9   to = proxmox_virtual_environment_vm.observability
10  id = "pve1/310"
11 }

```

Source: [platform/proxmox/terraform/homelab-guests/imports.tf](#), lines 1-11

With an empty state, a plain `resource` block would make OpenTofu propose creating VMs 300 and 310 again. The `import` blocks bind the resources to the existing guests by `node/vmid`. The first plan therefore shows an adoption, not a creation.

```

5 resource "proxmox_virtual_environment_vm" "homelab_services" {
6   vm_id      = 300
7   name       = "homelab-services"
8   node_name  = var.homelab_services_node
9   description = local.managed_description
10  tags       = ["homelab-services", "terraform", "ansible"]
11
12  protection          = true
13  on_boot             = true
14  started             = true
15  reboot_after_update = false
16  stop_on_destroy     = false
17  purge_on_destroy    = false
18  delete_unreferenced_disks_on_destroy = false

```

Source: [platform/proxmox/terraform/homelab-guests/main.tf](#), lines 5-18

Proxmox-level `protection` stops the guest being deleted from the UI or API. The three `*_on_destroy` settings make a destroy leave disks alone. `prevent_destroy` in the `lifecycle` block makes OpenTofu refuse any plan that would destroy the resource. The `lifecycle` block also ignores disks and cloud-init, which Ansible and the guest own.

```

9 locals {
10   lab      = yamldecode(file(var.lab_file))
11   rebuildable = { for name, guest in local.lab.guests : name => guest if try(guest.rebuild, false) }
12   selected   = { for name, guest in local.rebuildable : name => guest if contains(var.rebuild_guests, name) }
13   nodes     = toset([for guest in values(local.selected) : guest.node])

```

```

14
15     expected_confirmation = "REBUILD ${join(" ", sort(tolist(var.rebuild_guests)))} FROM ZERO"
16     description           = "Home Lab guest recreated by the rebuild-only root; Ansible configures
    it (L4+)"
17 }
18
19 resource "terraform_data" "rebuild_gate" {
20     count = length(var.rebuild_guests) > 0 ? 1 : 0
21
22     lifecycle {
23         precondition {
24             condition     = length(setsubtract(var.rebuild_guests, keys(local.rebuildable))) == 0
25             error_message = "rebuild_guests may only name guests with rebuild: true in inventory/
lab.yml."
26         }
27         precondition {
28             condition     = var.rebuild_confirmation == local.expected_confirmation
29             error_message = "rebuild_confirmation must equal the exact string \"REBUILD <sorted names>
FROM ZERO\"."
30         }
31     }
32 }

```

Source: [platform/proxmox/terraform/rebuild-guests/main.tf](#), lines 9–32

The rebuild root reads `inventory/lab.yml` directly and considers only guests marked `rebuild: true`. By default `rebuild_guests` is empty, so a plan is empty. To create anything, the operator must list guests by name **and** supply the confirmation `REBUILD <sorted names> FROM ZERO`; both are OpenTofu preconditions.

## 4.4 Operate

```

make -C platform/proxmox check           # offline: lint, tests, syntax
make -C platform/proxmox validate-ssh    # read-only
make -C platform/proxmox discover        # read-only
make -C platform/proxmox terraform-plan # import-first plan for 300/310
make -C platform/proxmox rebuild-guests-test # offline, mocked provider

```

There is no `apply` or `destroy` Make target anywhere. Applying a reviewed plan is an operator action, done in the local devcontainer. The operator reviews the plan and supplies the existing local approval; GitHub never plans or applies.

### WARNING

`qm set <vmid> --nameserver ...` on a **stopped** cloud-init guest regenerates its SSH host keys on the next boot. Expect a host-key change and verify the new key through `qm guest exec <vmid> -- /usr/bin/ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub` before the operator updates the pin.

## 4.5 Recover

A node is rebuilt with rungs L0–L2 of the [rebuild ladder](#). The rungs are: an answer-file ISO install, `pve_host_config` for jobs, onboot and startup, and a manual `pvecmd add`. Guests come back from `synology-backup` with `qmrestore` to a spare VMID, all NICs link-down, before anything is swapped in. VM 290 (EVE-NG) is only ever restored from `vzdump`.

**DANGER — DESTRUCTIVE STAGE**

Rebuilding a node discards its local guests. `pvecmd add`, ISO installs and any restore over a live VMID are manual operator steps. No Make target or workflow performs them.

## 5 NAS and network

### 5.1 Architecture

The Synology the NAS (`nas1`, 192.168.0.20) has two jobs here. It is the backup target (NFS on Volume 1), and it is the authoritative DNS server for the private zone `lab.example.com`. Volume 1 is a single 12 TB disk with **no redundancy**, so losing that disk loses every archive at once. Volume 2 is a RAID1 mirror whose Disk 3 has failed SMART and is scheduled for replacement.

D5 approves a WD Red Plus 4 TB CMR replacement in bay 3. The [attended replacement runbook](#) requires a fresh healthy Disk 4 result and a verified independent backup, then DSM Repair and supported post-rebuild checks. RAID1 parity scrubbing is unavailable in DSM; use the actual filesystem's offered scrub operation or record it unsupported and verify SMART/files instead. Volume 1's larger structural risk is addressed separately by adding a 12 TB+ CMR disk in empty bay 1 and converting its Basic pool to RAID1, never Basic to SHR.

DNS is split-horizon **without** the router. The the router must never be authoritative for, or forward, `lab.example.com`: doing so takes the operator's home-automation hub offline. Each guest therefore gets a systemd-resolved routing domain (`Domains=~lab.example.com`, `DNS=192.168.0.20`). Only private-zone queries go to the NAS; everything else stays on the the router, including its IPv6 resolver.

### 5.2 Files

| File                                                                      | Purpose                                         |
|---------------------------------------------------------------------------|-------------------------------------------------|
| <a href="#">platform/nas/ansible/playbooks/05-synology-pre-flight.yml</a> | Read-only NFS reachability and throughput probe |
| <a href="#">platform/nas/ansible/roles/synology_nfs/</a>                  | Preflight and gated storage registration        |
| <a href="#">platform/nas/docs/synology-storage-health.md</a>              | Disk and pool health, measured options          |
| <a href="#">observability/config/prometheus/file_sd/dns.yml</a>           | The only DNS target that may be probed          |

### 5.3 Key code

```
8 # `keep-all=1` is what makes this storage a write-only archive from Proxmox's
9 # point of view: no retention policy, so nothing is ever pruned automatically.
10 synology_nfs_prune_policy: "keep-all=1"
```

Source: [platform/nas/ansible/roles/synology\\_nfs/defaults/main.yml](#), lines 8–10

This one value makes the NFS storage append-only from Proxmox's point of view. With no retention policy, Proxmox never prunes an archive. The same value is repeated on every scheduled job by `pve_host_config`, and a test asserts that no job prunes.

```
1 ---
2 # Authoritative split-horizon DNS check for the private zone.
3 #
4 # ONLY the Synology at 192.168.0.20 is authoritative for lab.example.com.
5 #
6 # Do NOT re-add 192.168.0.1:53 here. The the router is deliberately NOT configured to
7 # serve this zone: making it authoritative takes the home-automation hub offline.
8 # Guests reach the private zone through a systemd-resolved routing domain
9 # (/etc/systemd/resolved.conf.d/lab-dns.conf) that sends only lab.example.com
```

```
10 # to 192.168.0.20, while general resolution stays on the the router. Probing the the router
11 # with the dns_lab module asserts a requirement this architecture rejects and
12 # fires DnsResolutionFailed and EndpointDown permanently.
13 - targets:
14   - 192.168.0.20:53
15   labels:
16     zone: lab.example.com
```

Source: [observability/config/prometheus/file\\_sd/dns.yml](#), lines 1-16

The comment records a decision that is easy to undo by mistake. Probing the the router for the lab zone would look like thorough monitoring, but it asserts a behaviour the architecture forbids. It would fire `DnsResolutionFailed` permanently.

## 5.4 Operate

```
make -C platform/nas check          # offline
make -C platform/proxmox nas-preflight # read-only NFS probe from all nodes
ssh nas                             # port 2258, user labadmin; sudo needs a password
```

## 5.5 Recover

The NAS is rung L0 and is entirely manual. The steps are: restore the DSM configuration backup, recreate the shares (`proxmox-cluster-backups` on Volume 1; `Media` and `media-backups` on Volume 2), recreate the NFS export and the DNS zone, and pin the SSH host key under `HostKeyAlias nas`. DSM has no provider, and its admin credentials are the operator's.

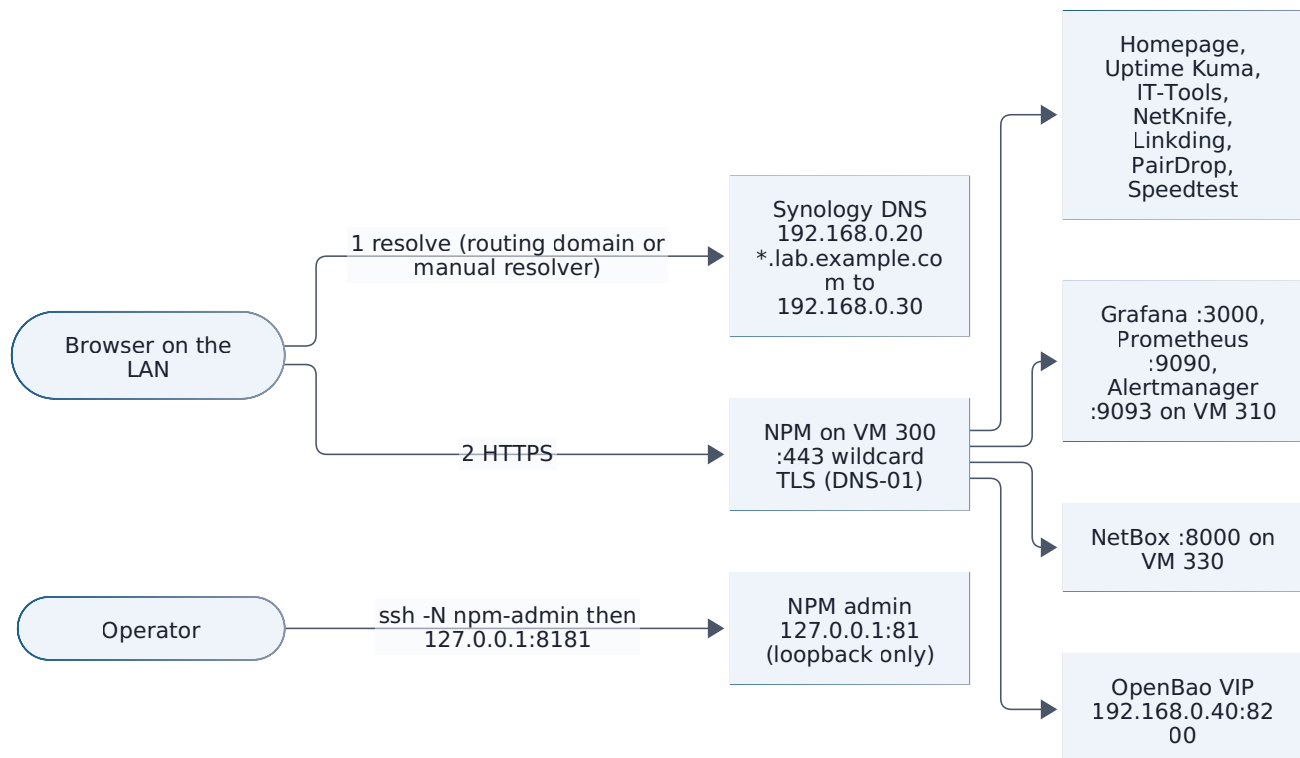
### WARNING

Never “simplify” guest DNS by pointing guests wholesale at 192.168.0.20. If the NAS is down, guests would lose all name resolution instead of only the private zone.

## 6 Homelab services (VM 300)

### 6.1 Architecture

VM 300 `homelab-services` runs on `pve2` at `192.168.0.30` (Ubuntu 24.04, 4 vCPU, 6 GiB, 120 GiB on `local-lvm`). It runs eight independent Compose projects with nine containers. Every image is digest-pinned and has an explicit memory limit and a health check. Nginx Proxy Manager terminates TLS for every private site, including the observability UIs on VM 310 and NetBox on VM 330. Its admin UI is bound to `127.0.0.1:81` and reached only through an SSH tunnel.



**Figure 8.** A browser request to a private site: DNS, TLS at NPM, then the backend

Diagram source: <docs/walkthrough/diagrams/08-vm300-request-path.mmd>

### 6.2 Files

| File                                                                                 | Purpose                                                              |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <code>services/homelab-services/ansible/playbooks/</code>                            | Guest config (30), preflight (31), one-app deploy (40), backups (50) |
| <code>services/homelab-services/ansible/roles/homelab_services/tasks/main.yml</code> | Renders exactly one Compose project; NPM route ownership gate        |
| <code>services/homelab-services/config/templates/stage-d/compose/</code>             | One <code>compose.yaml</code> per application                        |
| <code>services/homelab-services/config/templates/stage-d/images.lock.json</code>     | Image digests                                                        |
| <code>services/homelab-services/config/templates/stage-g/backup.py</code>            | Application-aware backup                                             |

| File                                                                   | Purpose                                        |
|------------------------------------------------------------------------|------------------------------------------------|
| <code>services/homelab-services/config/uptime-kuma/monitors.yml</code> | Declared monitors, compared with the live dump |

## 6.3 Key code

```

63 def backup_sqlite(source: Path, destination: Path) -> str:
64     if not source.is_file():
65         raise RuntimeError(f"required SQLite database is missing: {source}")
66
67     source_uri = f"file:{source}?mode=ro"
68     with (
69         sqlite3.connect(source_uri, uri=True) as source_db,
70         sqlite3.connect(destination) as destination_db,
71     ):
72         source_db.backup(destination_db)
73         result = destination_db.execute("PRAGMA integrity_check").fetchone()
74
75     if result != ("ok",):
76         raise RuntimeError(f"SQLite integrity check failed for {source}")
77     destination.chmod(0o600)
78     return "ok"

```

Source: `services/homelab-services/config/templates/stage-g/backup.py`, lines 63–78

SQLite files are never copied while the app is running. The source is opened read-only through a URI, and SQLite's online backup API writes a consistent copy. `PRAGMA integrity_check` must return exactly `ok` or the run fails. Uptime Kuma 2 uses MariaDB, not SQLite, so it gets a transactional `mariadb-dump` instead.

## 6.4 Operate

```

make -C platform/proxmox homelab-preflight          # read-only
make -C platform/proxmox homelab-service SERVICE=homepage \
    EXTRA_JSON='{ "homelab_allow_service_deploy":true,"homelab_service_deploy_confirmation":"DEPLOY
    homepage ON homelab-services"}'
ssh -N npm-admin # then browse http://127.0.0.1:8181 for the NPM admin UI

```

One run reconciles one application. The Homepage card policy is tested: deployed cards need both an `href` and a `siteMonitor`, and planned entries have neither.

## 6.5 Recover

The backup timer runs at 02:45 and writes to `/srv/homelab-backups/staging`. The 03:15 `vzdump` of VM 300 carries that off the guest. A restore goes to an isolated copy (protected VMID 399, NIC link-down). The SQLite files are copied back with the app stopped, and the Kuma dump is replayed into its MariaDB. This path was proven end to end before VM 300 was set to `onboot=1`.

## 7 NetBox (VM 330)

### 7.1 Architecture

VM 330 netbox runs on pve3 at 192.168.0.50 with six containers. UFW admits port 8000 only from VM 300, and NPM serves `netbox.lab.example.com` and `ipam.lab.example.com`. The database starts empty; the rebuild hint from the destroyed pve2 snapshot is not a backup. Its OpenTofu root ([services/netbox/terraform/proxmox-guest/](#)) is separate from the VM 300/310 root on purpose. A drift plan for the adopted guests can therefore never also propose creating VM 330.

### 7.2 Files

| File                                                                     | Purpose                                                      |
|--------------------------------------------------------------------------|--------------------------------------------------------------|
| <a href="#">services/netbox/ansible/playbooks/</a>                       | Preflight (30), guest config (31), deploy (40), backups (50) |
| <a href="#">services/netbox/config/templates/stage-a/compose.yaml.j2</a> | NetBox Compose project                                       |
| <a href="#">services/netbox/config/templates/stage-b/backup.py</a>       | pg_dump plus media, hashed, never pruned                     |
| <a href="#">services/netbox/scripts/netbox_seed_csv.py</a>               | Offline CSV seed built from inventory                        |

### 7.3 Key code

```

66 def dump_postgres(destination: Path) -> None:
67     """Stream a single-transaction pg_dump straight into a gzip file."""
68     command = [
69         DOCKER,
70         "exec",
71         POSTGRES_CONTAINER,
72         "pg_dump",
73         "--username=netbox",
74         "--dbname=netbox",
75         "--format=plain",
76         "--no-owner",
77         "--no-privileges",
78         "--serializable-deferrable",
79     ]
80     with (
81         subprocess.Popen(command, stdout=subprocess.PIPE, stderr=subprocess.PIPE) as process,
82         gzip.open(destination, "wb", compresslevel=6) as output,
83     ):
84         if process.stdout is None:
85             raise RuntimeError("pg_dump did not open stdout")
86         shutil.copyfileobj(process.stdout, output, length=1024 * 1024)
87         process.stdout.close()
88         returncode = process.wait()
89     if returncode != 0:
90         raise RuntimeError(f"NetBox pg_dump failed with exit code {returncode}")
91     if destination.stat().st_size == 0:
92         raise RuntimeError("NetBox pg_dump produced an empty archive")

```

Source: [services/netbox/config/templates/stage-b/backup.py](#), lines 66–92

The dump runs inside the Postgres container as a single `--serializable-deferrable` transaction. It gets a consistent snapshot without blocking writers. Output streams straight into gzip without touching disk uncompressed. A non-zero exit or an empty file is an error, never a silently truncated backup.

## 7.4 Operate and recover

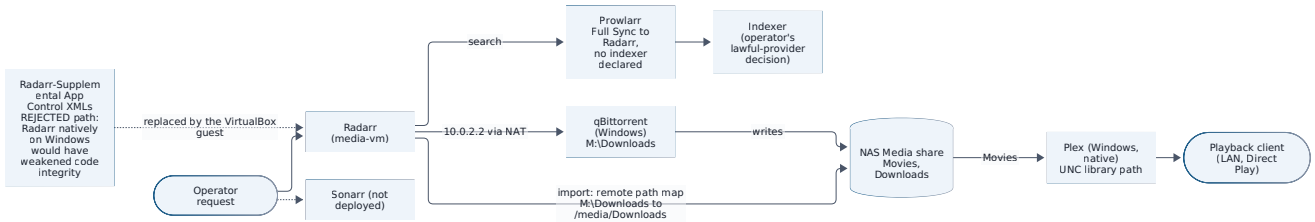
```
make -C platform/proxmox netbox-preflight      # read-only  
make -C platform/proxmox netbox-deploy EXTRA_JSON='{"netbox_allow_deploy":true,"netbox_deploy_confirmation":"DEPLOY NETBOX ON netbox"}'
```

The backup runs at 03:45 and the 04:45 vzdump carries it off the guest. To restore, load the dump into an isolated VM 330 copy with an empty database, then compare object counts. That isolated restore and `onboot=1` are still gated; see `MEMORY.md` (not published).

## 8 Media (workstation)

### 8.1 Architecture

workstation stays Windows 11 Home. Plex Media Server and qBittorrent run natively. Radarr, Prowlarr, Recyclarr, the exportarr exporters and Alloy run in Docker inside `media-vm`, a VirtualBox guest. The library lives on the NAS `Media` share. Sonarr is not deployed. Running Radarr natively on Windows would have needed App Control policy changes, which were rejected. The VirtualBox guest replaced that path.



**Figure 9.** Media request flow: Radarr and Prowlarr in the guest, qBittorrent and Plex on Windows, library on the NAS

Diagram source: `docs/walkthrough/diagrams/09-media-flow.mmd`

### 8.2 Files

| File                                                  | Purpose                                                                                          |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>services/media/config/compose.yaml</code>       | Digest-pinned container definitions                                                              |
| <code>services/media/config/cloud-init/</code>        | Guest first boot (L0)                                                                            |
| <code>services/media/ansible/roles/</code>            | <code>media_host</code> , <code>arr_stack</code> , <code>media_backup</code> , <code>plex</code> |
| <code>services/media/terraform/arr/</code>            | *arr settings through the devopsarr providers                                                    |
| <code>services/media/config/recyclarr/</code>         | Quality profiles (ADR-0004)                                                                      |
| <code>services/media/scripts/plex_libraries.py</code> | Plex library plan (plan by default)                                                              |

### 8.3 Key code

```
1 ---
2 name: lux-media
3 # media-vm containers. radarr and prowlarr are the live definitions
4 # recovered from the guest bootstrap (unchanged). Everything else was added in
5 # Phase 3 and is deployed by the arr_stack role under its gate.
6 # API keys are operator-installed env files under /etc/media/secrets (0600
7 # root); Docker reads them as root, so container users never need access.
8 services:
9   radarr:
10     image: lscr.io/linuxserver/
11     radarr:latest@sha256:adb6c09d6b729ea5e642c99cea35af72702ef476bf4763f153299ac5db9f0b4f
12     container_name: radarr
13     restart: unless-stopped
14     environment:
15       PUID: "1000"
16       PGID: "1000"
17       TZ: America/Los_Angeles
18     ports:
```

```
18     - "7878:7878"
19     volumes:
20     - /srv/radarr/config:/config
21     - /mnt/media:/media
22     tmpfs:
23     - /run/radarr-temp:uid=1000,gid=1000,mode=0700
24     cap_drop:
25     - ALL
26     cap_add:
27     - SETGID
28     - SETUID
29     mem_limit: 1g
30     mem_reservation: 256m
31     cpus: 1.0
32     pids_limit: 512
```

Source: `services/media/config/compose.yaml`, lines 1-32

Each image is pinned by digest even when its tag is `latest`, so a rebuild gets the same bytes. Containers drop every capability and add back only `SETUID` and `SETGID`, which the LinuxServer images need to switch users. Each one has hard memory, CPU and PID limits. API keys are never in the file; they come from root-owned env files under `/etc/media/secrets`.

## 8.4 Operate and recover

```
make -C services/media check           # offline
make -C services/media preflight       # read-only
make -C services/media sync-quality-profiles # Recyclarr --preview unless gated
```

Metrics are pushed from the guest to VM 310 ([ADR-0006](#)). The guest sits behind VirtualBox NAT with loopback-only forwards, so VM 310 cannot scrape it. workstation also sleeps, so alerts treat an absent series as expected. Radarr's and Prowlarr's own scheduled backups are copied nightly, append-only, to the NAS `media-backups` share. Plex's backup task is behind its own gate.

### WARNING

Nothing always-on may live on workstation: it sleeps after five hours on AC. Do not stop or reconfigure the Docker Desktop `radarr` container there; that decision is the operator's.

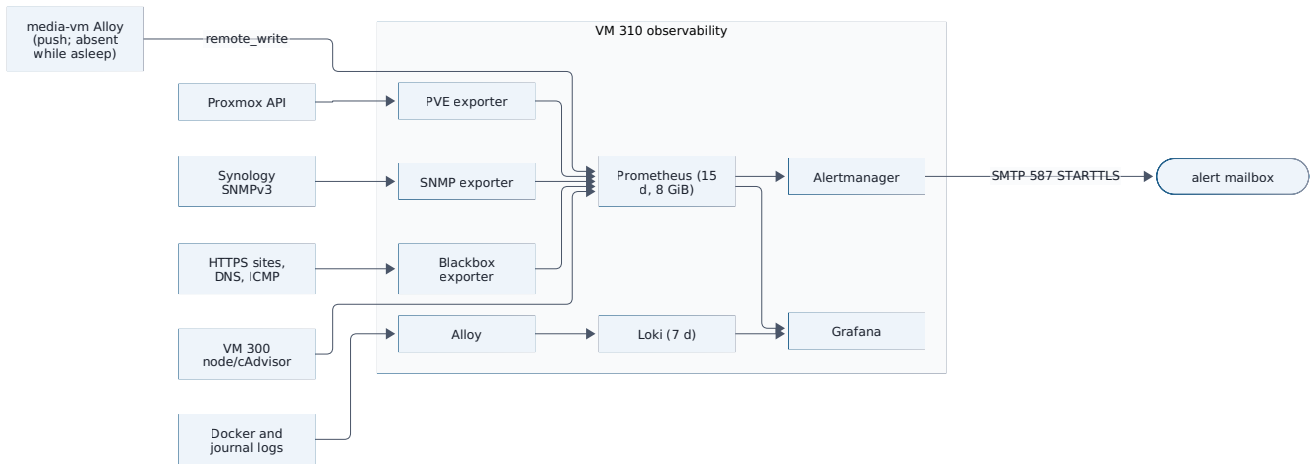
## 8.5 Music Library

`services/music-library/` (not published) is a local, on-demand music library for workstation. It has a PySide6 desktop app and a FastAPI web companion for other devices on the LAN. It is its own Python project with its own CI job. Its architecture, legal boundaries and operation are documented in `services/music-library/README.md` (not published), which is the single source for that component.

## 9 Observability (VM 310)

### 9.1 Architecture

VM 310 runs the central stack on pve1 NVMe: Prometheus (15 days, 8 GiB cap), Grafana, Loki (seven days), Alertmanager (SMTP through a dedicated mailbox), Alloy, and node, cAdvisor, blackbox, SNMP and PVE exporters. That is ten containers, each with a hard memory limit. The Synology is monitored over SNMPv3 read-only and never hosts observability storage. The PVE exporter uses a read-only API token and installs nothing on the hypervisors.



**Figure 10.** Metrics, logs and alerts: scrape and push paths into VM 310, alerts out by email

Diagram source: [docs/walkthrough/diagrams/10-observability-flow.mmd](#)

### 9.2 Files

| File                                                | Purpose                                                         |
|-----------------------------------------------------|-----------------------------------------------------------------|
| <code>observability/docker/compose.yaml</code>      | The ten-container stack                                         |
| <code>observability/config/prometheus/</code>       | Scrape config, file-based targets, alert rules                  |
| <code>observability/config/grafana/</code>          | Provisioned dashboards (Synology, Proxmox, fleet, availability) |
| <code>observability/scripts/backup.sh</code>        | TSDB snapshot plus Grafana DB and config archive                |
| <code>observability/scripts/restore-drill.sh</code> | Validates an archive without touching live data                 |
| <code>observability/ansible/playbooks/</code>       | Bootstrap (20), deploy (40), dashboard update (41)              |

### 9.3 Key code

```
12 # Prometheus binds the LAN address, not loopback (see compose.yaml:
13 # ${OBSERVABILITY_IP}:9090:9090). Reading the address from the same .env that
14 # Compose uses keeps the two definitions from drifting. A hardcoded
15 # http://127.0.0.1:9090 here silently failed every nightly run with curl exit 7
16 # and produced no archive at all between 2026-08-24 and 2026-09-06.
17 observability_ip="$(sed -n 's/^OBSERVABILITY_IP=//p' "${stack_root}/docker/.env" 2>/dev/null | tail -1)"
18 readonly prometheus_url="http://${observability_ip:-192.168.0.31}:9090"
```

Source: [observability/scripts/backup.sh](#), lines 12-18

Prometheus binds the LAN address, not loopback. A hardcoded `127.0.0.1` made every nightly backup fail for two weeks with no alert. The script now reads the same `.env` file Compose uses, so the two cannot drift apart again.

```

76 # The script-wide `umask 077` above is correct for the archive, which carries
77 # configuration, but it also made this metric file 0600 root:root. The
78 # node_exporter container runs as `nobody` and could not read it, so the
79 # textfile collector reported node_textfile_scrape_error=1 and
80 # homelab_observability_backup_last_success_timestamp_seconds NEVER EXISTED in
81 # Prometheus. A freshness alert written against it would have been vacuous: it
82 # would evaluate against no series, never fire, and look exactly like health.
83 # Discovered 2026-09-19. The file contains a single Unix timestamp and no
84 # secret, so 0644 is appropriate. Do not "fix" this by relaxing the umask.
85 metric_tmp="${metric_dir}/homelab_observability_backup.prom.tmp"
86 printf 'homelab_observability_backup_last_success_timestamp_seconds %s\n' "$(date +%s)" >"$
{metric_tmp}"
87 chmod 0644 "${metric_tmp}"
88 mv "${metric_tmp}" "${metric_dir}/homelab_observability_backup.prom"

```

Source: [observability/scripts/backup.sh](#), lines 76-88

The freshness metric is written atomically (temporary file, then `mv`), and it is made world-readable on purpose. The comment records why. The first version was `0600`, the exporter runs as `nobody`, and the metric never reached Prometheus. The freshness alert built on it could never fire.

```

51 - alert: ObservabilityBackupMetricMissing
52   expr: absent(homelab_observability_backup_last_success_timestamp_seconds)
53   for: 30m
54   labels:
55     severity: critical
56   annotations:
57     summary: "Backup freshness metric has disappeared"
58     description: >-
59       homelab_observability_backup_last_success_timestamp_seconds is absent
60       from Prometheus, so ObservabilityBackupStale cannot fire and backup
61       failures are invisible. Check the textfile file mode
62       (/var/lib/node_exporter/textfile, must be world-readable), the
63       node_exporter bind mount, and node_textfile_scrape_error.
64
65   # Guard 2: catch the collector failing for any reason - permissions, a
66   # malformed .prom file, or a broken mount - rather than only noticing when
67   # a specific series vanishes.
68 - alert: NodeTextfileCollectorFailing
69   expr: node_textfile_scrape_error > 0
70   for: 15m
71   labels:
72     severity: warning
73   annotations:
74     summary: "node_exporter textfile collector is failing"
75     description: >-
76       node_textfile_scrape_error is non-zero, so at least one .prom file
77       cannot be read. Any alert depending on a textfile metric may be
78       silently inert. Check `docker logs` for the node-exporter container.

```

Source: [observability/config/prometheus/rules/observability.yml](#), lines 51-78

These two rules guard the guard. `absent()` fires if the backup metric disappears.

`node_textfile_scrape_error > 0` fires if any textfile metric cannot be read. Together they stop the staleness alert from silently becoming vacuous again. They must never be deleted.

## 9.4 Operate

```
make -C observability check      # offline, includes promtool
make -C observability validate   # read-only runtime checks
make -C observability restore-drill ARCHIVE=/srv/observability/backups/<file>
```

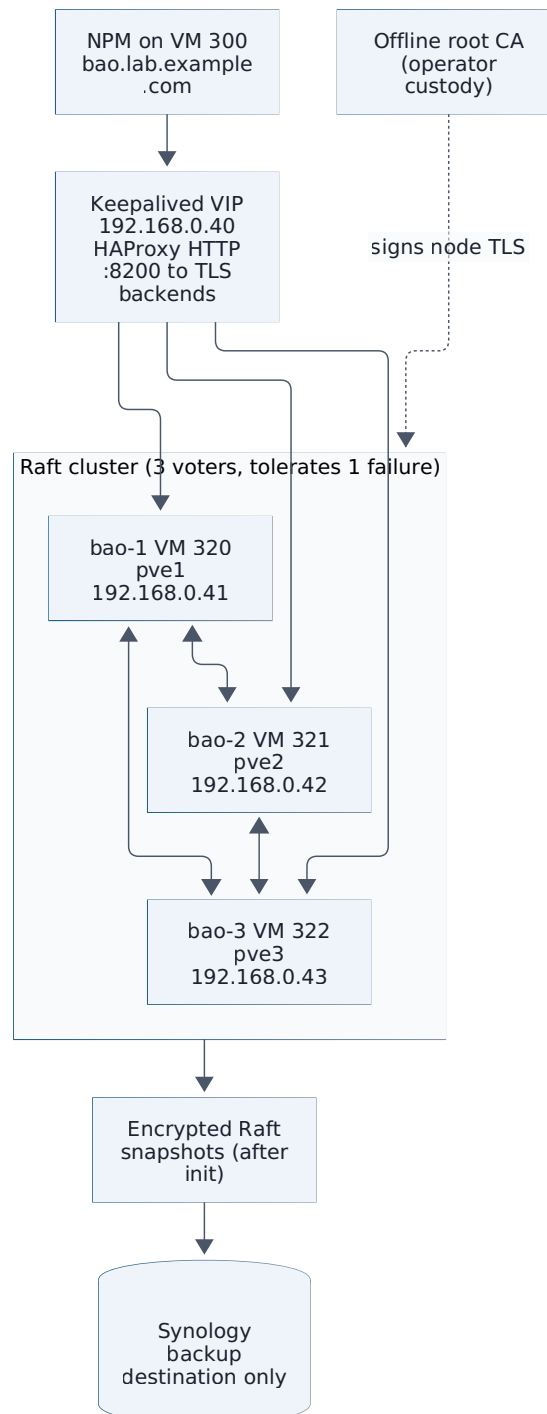
## 9.5 Recover

The backup runs at 03:30 into `/srv/observability/backups` (root-only, so use `sudo -n bash -c '...'` for globbing). The 04:15 `vzdump` carries it off the guest. Loki history is disposable. Rebuilding means: the guest from L3, the stack from the gated deploy, and Prometheus and Grafana state from the latest archive. VM 310 stays `onboot=0` until its restore and onboot gates pass.

## 10 Secrets — OpenBao

### 10.1 Architecture

Three OpenBao 2.6.1 voters (VMs 320–322), one on each node, form a Raft cluster. Keepalived holds the VIP 192.168.0.40, where HAProxy terminates HTTP on port 8200 and re-encrypts to each node over TLS. NPM publishes `bao.lab.example.com`. TLS is valid everywhere, and the cluster is deliberately **uninitialized**. Initialization, unsealing and rekeying are attended ceremonies that no automation performs.



**Figure 11.** OpenBao voters, the Keepalived VIP and the HAProxy front end

Diagram source: <docs/walkthrough/diagrams/11-openbao-topology.mmd>

## 10.2 Files

| File                                                                                   | Purpose                                                                                                          |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <a href="#">security/secrets-openbao/ansible/playbooks/</a>                            | Prepare (10), install (20), TLS (30), HA (40), validate (50), post-init (60/65), backup (70/90), monitoring (80) |
| <a href="#">security/secrets-openbao/openbao/policies/</a>                             | Least-privilege policies, applied after init                                                                     |
| <a href="#">security/secrets-openbao/scripts/raft_snapshot.py</a>                      | Encrypted (age) Raft snapshots                                                                                   |
| <a href="#">security/secrets-openbao/docs/bootstrap.md</a> , <a href="#">unseal.md</a> | The attended initialization and unseal ceremony                                                                  |
| <a href="#">security/secrets-openbao/docs/restore-runbook.md</a>                       | Isolated restore                                                                                                 |

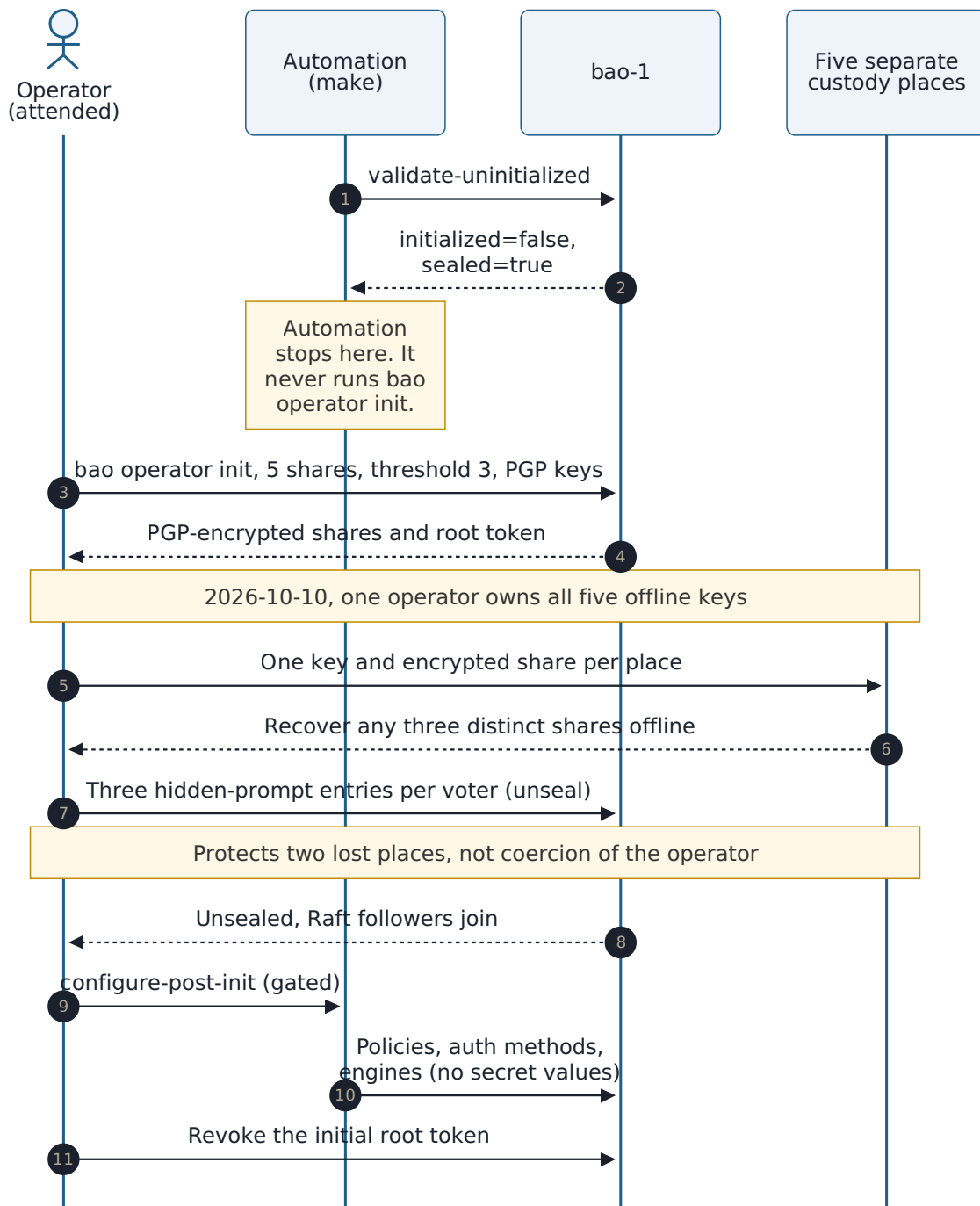
## 10.3 Key code

```

90 - name: Query OpenBao initialization state over verified TLS
91   ansible.builtin.uri:
92     url: "https://{{ inventory_hostname }}.{{ openbao_domain }}:8200/v1/sys/init"
93     method: GET
94     ca_path: /etc/openbao/tls/ca.pem
95     return_content: true
96     status_code: 200
97   register: openbao_cluster_init_status
98   changed_when: false
99
100 - name: Stop before initialization and require an attended ceremony
101   ansible.builtin.assert:
102     that:
103       - not (openbao_cluster_init_status.json.initialized | bool)
104     fail_msg: >-
105       This validation gate expects an uninitialized cluster. If initialization
106       was intentional, use the post-init validation stage instead.
107     success_msg: "TLS is valid and OpenBao remains uninitialized. Stop for the attended
    ceremony."
```

Source: [security/secrets-openbao/ansible/roles/openbao\\_cluster/tasks/main.yml](#), lines 90–107

The validation stage proves two things over verified TLS (the CA path is pinned): every node answers, and `initialized` is still false. If someone initialized the cluster, this stage fails rather than carrying on. The post-init stages are separate playbooks with separate gates.

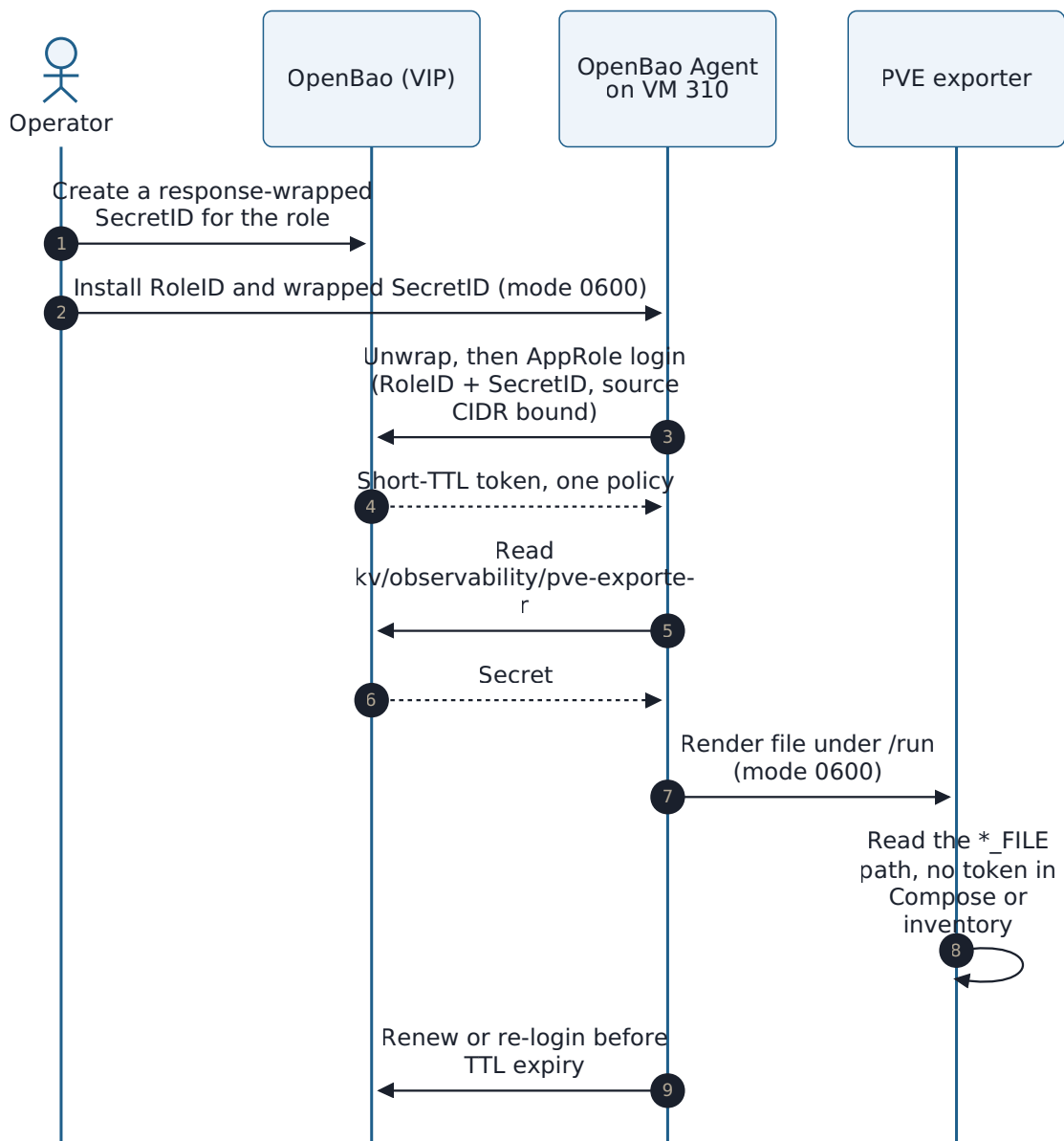


**Figure 12.** The attended initialization ceremony: automation stops before init

Diagram source: <docs/walkthrough/diagrams/12-openbao-init-ceremony.mmd>

### **DANGER — DESTRUCTIVE STAGE**

Never run `bao operator init`, `unseal` or `rekey` from automation, CI or an agent. The guard hook and the Codex rules block the whole `bao operator` family. D4 schedules the attended ceremony for **2026-10-10**. One operator owns five PGP keys with separate passphrases in five custody places, threshold three. This protects against losing any two places, not coercing that operator. The [ceremony runbook](#) gives offline key generation, fingerprint verification, operator commands and separate post-init gates. The day before requires a verified `vzdump` archive of each VM 320–322. The initial root token is revoked after replacement access works.



**Figure 13.** After initialization: an OpenBao Agent renders a secret to a file for the PVE exporter

Diagram source: <docs/walkthrough/diagrams/13-openbao-approle.mmd>

The first planned integration is the read-only PVE exporter token. It uses a response-wrapped AppRole SecretID, an Agent that renders the secret to a `0600` file under `/run`, and a `*_FILE` variable in Compose. No token ever appears in Git, inventory or Compose.

## 10.4 Operate and recover

```

make -C security/secrets-openbao check # offline
make -C security/secrets-openbao validate-uninitialized # read-only: TLS valid, still uninitialized
make -C security/secrets-openbao recovery-bundle # secret-free configuration bundle
  
```

Before initialization there is no data to lose. The hosts are rebuilt from L3 and the gated prepare, install, TLS and HA stages. After initialization, encrypted Raft snapshots are restored into a protected VM on an isolated bridge by following the restore runbook. The age identity that decrypts them lives outside both OpenBao and the NAS.

# 11 AAA — FreeRADIUS

---

[security/aaa-freeradius/](#) holds discovery and a read-only preflight playbook only. Nothing is deployed. The existing `aaa-ldap-radius` VirtualBox guest on the operator's Mac has not been inspected through trusted access. PiVPN has no RADIUS listener. The plan and its required inputs are in [security/aaa-freeradius/docs/aaa-deployment.md](#). RADIUS and LDAP must never be exposed to the Internet.

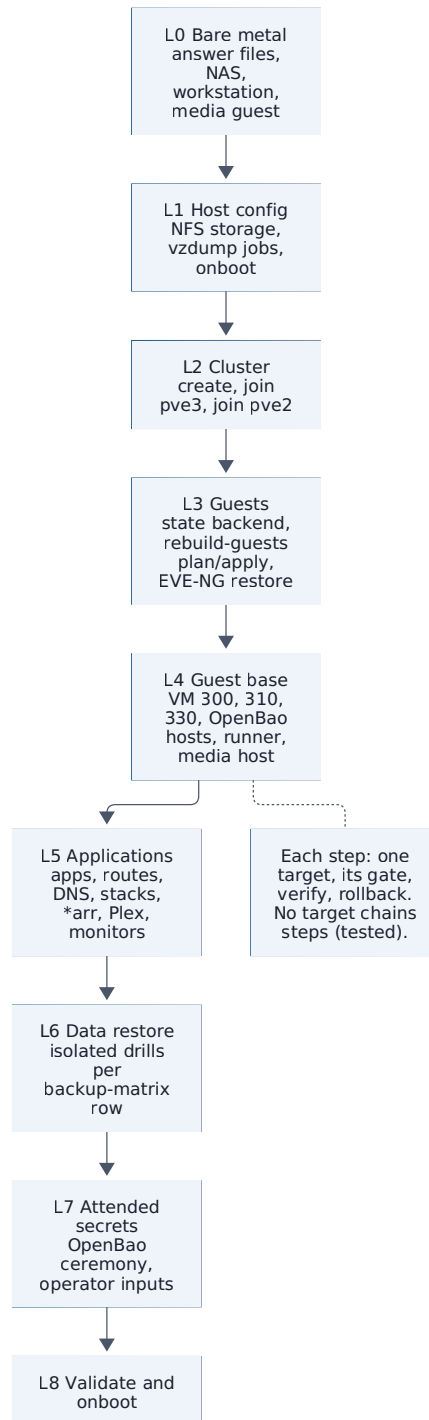
## 12 Rebuild ladder

The whole lab can be rebuilt from bare metal by climbing nine rungs, L0 to L8. Each step is one Make target or one manual action, with its own gate, verify command and rollback. The ladder is data ([docs/runbooks/rebuild-ladder.yml](#)), and both the runbook table and `make rebuild-plan` are generated from it:

```
10 rungs:
11   - id: L0
12     title: Bare metal and base installs
13     steps:
14       - id: l0-proxmox-answer-files
15         component: platform/proxmox
16         make: rebuild-l0-answer-files NODE=<node> WRITE=1
17         gate: {kind: local, reason: "Writes git-ignored files only; secrets come from the
operator's environment"}
18         inputs: ["lab.yml nodes.<node>.install.{fqdn,disk_filter,interface_filter}",
PVE_ROOT_PASSWORD_HASH, PVE_ROOT_SSH_KEYS_FILE, PVE_MAILTO]
19         duration: 5 min
20         verify: proxmox-auto-install-assistant validate-answer artifacts/autoinstall/<node>.toml
21         rollback: Discard the generated file; nothing outside the controller changed.
```

Source: [docs/runbooks/rebuild-ladder.yml](#), lines 10–21

A step carries its component, its single Make target, the kind of gate (`local`, `read-only`, `manual`, or a `{flag, confirmation}` pair), the operator inputs it needs, a verify command and a rollback. Tests check that every target exists and that no target chains rungs. They also check that every gate flag and confirmation string really appears in the source.



**Figure 14.** Rungs L0 to L8, from bare metal to validated onboot

Diagram source: <docs/walkthrough/diagrams/06-rebuild-ladder.mmd>

The runbook is included here unchanged. It is the single source for the procedure:

## 12.1 Rebuild from zero (L0 → L8)

How to rebuild the whole lab from bare metal with this repository plus the backups in <docs/reference/backup-matrix.yml>. Climb one rung at a time:

- Every step is **one** Make target or one manual action. No target chains steps (tested), and there is no aggregate target of any name.

- Run the step's `verify` before the next step. Stop on any surprise.
- Gates are passed as `EXTRA_JSON`, for example `make -C platform/proxmox rebuild-l1-host-config EXTRA_JSON='{ "pve_host_config_allow":true,"pve_host_config_confirmation":"L1 CONFIGURE pve pve1 pve2"}'`. An agent prepares these strings (skill `run-gated-stage`); only the operator supplies them.
- `make rebuild-plan` prints this ladder and checks local prerequisites offline (tools, `null` inputs in `inventory/lab.yml`, `backend.hcl`, which environment variables are set — never their values).

Before starting: read [docs/reference/lessons-learned.md](#) (cloud-init regenerates host keys on stopped guests; `/etc/pve` symlinks; the the router must never own `lab.example.com`), [docs/reference/operator-inputs.md](#) (L7) and [docs/reference/coverage-matrix.yml](#) (what is declared, restored or manual).

Known gaps that affect a rebuild are listed in `MEMORY.md`. In particular: live OpenTofu state is still only in the Phase 1 source archive (ADR-0003), and NFS storage registration has a flag-only gate.

### 12.1.1 The ladder

The table is generated from [docs/runbooks/rebuild-ladder.yml](#) (`python3 tools/rebuild/rebuild_plan.py --markdown`); edit the YAML, not the table. A test fails if they differ.

#### 12.1.1.1 L0 — Bare metal and base installs

| Step                                 | Command                                                                                 | Gate                                                                                  | Inputs                                                                                                                                                                                                                                                | Duration              | Verify                                                                                                                                                    | Rollback                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <code>l0-proxmox-answer-files</code> | <code>make -C platform/proxmox rebuild-l0-answer-files NODE=&lt;node&gt; WRITE=1</code> | local:<br>Writes git-ignored files only; secrets come from the operator's environment | <code>lab.yml</code> nodes.. <code>install</code> .<br>{ <code>fqdn</code> , <code>disk_filter</code> , <code>interface_filter</code> };<br><code>PVE_ROOT_PASSWORD_HASH</code> ;<br><code>PVE_ROOT_SSH_KEYS_FILE</code> ;<br><code>PVE_MAILTO</code> | 5 min                 | <code>proxmox-auto-install-assistant validate-answer artifacts/autoinstall/.toml</code> ler changed.                                                      | Discard the generated file; nothing outside the control-ler changed. |
| <code>l0-proxmox-install</code>      | manual                                                                                  | manual:<br>ISO preparation and console install are operator steps by design           | PVE 9 ISO; answer file from <code>l0-proxmox-answer-files</code>                                                                                                                                                                                      | 30 min per node       | Compare the console host-key fingerprint with <code>lab.yml</code> nodes.. <code>host_fingerprint</code> , then <code>ssh &lt;alias&gt; pveversion</code> | Reinstall the node; no guest exists yet.                             |
| <code>l0-nas</code>                  | manual                                                                                  | manual:<br>DSM has no provider; NAS admin credentials are the operator's              | DSM configuration backup; <code>lab.yml</code> nas                                                                                                                                                                                                    | 2 h plus RAID re-sync | <code>ssh nas</code> succeeds with the pinned key; shares <code>proxmox-cluster-backups</code> , <code>Media</code> and <code>media-backups</code> exist  | Restore the DSM configuration backup.                                |

| Step           | Command | Gate                                                               | Inputs                                                                | Duration           | Verify                                        | Rollback                                                               |
|----------------|---------|--------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------|-----------------------------------------------|------------------------------------------------------------------------|
| l0-workstation | manual  | manual: workstation stays Windows and is never wiped by automation | Operator's own Windows backup                                         | operator-dependent | ssh workstation host-name returns workstation | Not applicable; nothing in this repository changes Windows.            |
| l0-media-guest | manual  | manual: VirtualBox lifecycle on workstation is attended            | services/media/config/cloud-init/user-data.yml.tmpl; admin public key | 30 min             | make -C services/media preflight              | Delete only the new VM; the retained media-vm disks are never touched. |

#### Operator actions:

- `l0-proxmox-install`: `proxmox-auto-install-assistant prepare-iso <pve.iso> -fetch-from iso -answer-file .toml`; boot the node from it; pin the host key after verifying it at the console.
- `l0-nas`: Restore DSM, recreate shares (proxmox-cluster-backups on Volume 1, Media and media-backups on Volume 2), the plex-reader and media-backup accounts, NFS export for the cluster, and the split-horizon DNS zone from lab.yml dns\_records.
- `l0-workstation`: Restore Windows from the operator's backup, then install VirtualBox, Plex Media Server and qBittorrent. Plex claim and the client-scoped firewall rule stay attended (L5).
- `l0-media-guest`: Create a 2 vCPU / 4 GiB VM with a 50 GiB data disk, NAT with a loopback-only SSH forward, fill the cloud-init template, build the seed with cloud-localds, attach it and boot.

#### 12.1.1.2 L1 — Host configuration

| Step           | Command                                                      | Gate                                                                                                         | Inputs                                                        | Duration | Verify                                                                                  | Rollback                                                                                         |
|----------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| l1-nfs-storage | <code>make -C platform/proxmox nas-pre-flight</code>         | allow_configure_nfs_storage (flag only)<br>— gap: Flag-only gate predates Phase 3; add a confirmation string | NAS NFS export for the cluster                                | 10 min   | <code>pvesh get /nodes//storage/synology-backup/status</code> on each node shows active | <code>pvesm remove synology-backup</code> (removes the storage definition only, never archives). |
| l1-host-config | <code>make -C platform/proxmox rebuild-l1-host-config</code> | <code>pve_host_config_allow + L1 CONFIGURE pve pve1 pve2</code>                                              | inventory/lab.yml<br>backup_jobs<br>and guest on-boot/startup | 5 min    | <code>pvesh get /cluster/backup</code> lists every lab.yml backup job                   | Disable a created job in the UI (never delete archives); <code>qm set -onboot 0</code> .         |

### 12.1.1.3 L2 — Cluster

| Step              | Command                                              | Gate                                                                             | Inputs                                                             | Duration | Verify                                               | Rollback                              |
|-------------------|------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------|----------|------------------------------------------------------|---------------------------------------|
| l2-create-cluster | <code>make -C platform/proxmox create-cluster</code> | <code>al-low_create_cluster + CREATE CLUSTER homelab ON pve1 192.168.0.11</code> | three nodes installed and configured                               | 10 min   | pvecm status on pve1 shows one node, quorate         | Reinstall pve1 (no guests exist yet). |
| l2-join-pve3      | <code>make -C platform/proxmox join-pve3</code>      | <code>al-low_join_pve3 + JOIN pve3 192.168.0.13 TO homelab</code>                | root password prompt answered at the console (pvecm add is manual) | 15 min   | pvecm status shows two nodes, quorate                | Reinstall pve3 and rejoin.            |
| l2-join-pve2      | <code>make -C platform/proxmox join-pve2</code>      | <code>al-low_join_pve2 + JOIN pve2 192.168.0.12 TO homelab</code>                | root password prompt answered at the console                       | 15 min   | pvecm status shows three nodes, three votes, quorate | Reinstall pve2 and rejoin.            |

### 12.1.1.4 L3 — Guests

| Step             | Command                                                                                             | Gate                                                                             | Inputs                                                                                                                                         | Duration        | Verify                                                                            | Rollback                                                                     |
|------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| l3-state-backend | manual                                                                                              | manual: State migration is the operator's (ADR-0003); agents never migrate state | D:/homelab-state mounted in devcontainer; encrypted NAS state copy or archived plaintext source; per-root TF_VAR_state_passphrase              | 30 min per root | tofu plan shows no changes after migration                                        | Keep using the archived local state; delete nothing until the plan is clean. |
| l3-guests-plan   | <code>make -C platform/proxmox rebuild-guests-plan REBUILD_VARS=' -var-file=rebuild.tfvars '</code> | <code>re-build_guests + REBUILD &lt;sorted guest names&gt; FROM ZERO</code>      | backend.hcl (ADR-0003); TF_VAR_proxmox_api_token; TF_VAR_state_passphrase; git-ignored rebuild.tfvars with names, confirmation and public keys | 10 min          | The saved plan creates only the named VMIDs with lab.yml MACs, off and protected. | Discard the plan file.                                                       |

| Step             | Command | Gate                                                                     | Inputs                                            | Duration | Verify                                                                          | Rollback                                                                                       |
|------------------|---------|--------------------------------------------------------------------------|---------------------------------------------------|----------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| l3-guests-apply  | manual  | manual: There is no apply target; the operator applies the reviewed plan | rebuild.tfplan from l3-guests-plan                | 20 min   | qm list on each node shows the new VMIDs stopped; qm config shows protection: 1 | Guests are protected and prevent_destroy is set; removing one is a separate attended decision. |
| l3-eveng-restore | manual  | manual: Restores overwrite a VMID; attended by design                    | latest verified VM 290 archive on synology-backup | 1 h      | qm config 290 matches the archive; boot with link_down on every NIC first       | Leave the restored VM stopped and protected.                                                   |

Operator actions:

- `l3-state-backend`: Follow docs/runbooks/state-migration.md in order (OpenBao, NetBox, homelab-guests import-first). Restore encrypted NAS state or re-adopt by import; require a no-change plan before any apply.
- `l3-guests-apply`: `tofu -chdir=platform/proxmox/terraform/rebuild-guests apply rebuild.tfplan`
- `l3-eveng-restore`: `qmrestore 290 -storage local on pve1`, then verify.

#### 12.1.1.5 L4 — Guest base configuration

| Step               | Command                                                     | Gate                                                                         | Inputs                                                       | Duration | Verify                                                  | Rollback                                         |
|--------------------|-------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------|----------|---------------------------------------------------------|--------------------------------------------------|
| l4-vm300-base      | <code>make -C platform/proxmox homelab-base</code>          | <code>homelab_allow_guest_config + CONFIGURE GUEST homelab-services</code>   | VM 300 host key verified out of band through the guest agent | 15 min   | <code>make -C platform/proxmox homelab-preflight</code> | Re-run with the previous commit's configuration. |
| l4-vm310-bootstrap | <code>make -C observability bootstrap</code>                | <code>observability_allow_bootstrap + BOOTSTRAP observability</code>         | VM 310 host key verified out of band                         | 15 min   | <code>make -C observability preflight</code>            | Re-run with the previous commit's configuration. |
| l4-vm330-base      | <code>make -C platform/proxmox netbox-base</code>           | <code>netbox_allow_guest_config + CONFIGURE GUEST netbox</code>              | VM 330 host key verified out of band                         | 15 min   | <code>make -C platform/proxmox netbox-preflight</code>  | Re-run with the previous commit's configuration. |
| l4-openbao-hosts   | <code>make -C security/secrets-openbao prepare-hosts</code> | <code>allow_prepare_openbao_hosts + PREPARE OPENBAO HOSTS 320 321 322</code> | VMs 320-322 host keys verified out of band                   | 20 min   | <code>make -C security/secrets-openbao preflight</code> | Re-run with the previous commit's configuration. |

| Step          | Command                               | Gate                                      | Inputs                                                   | Duration | Verify                           | Rollback                                                            |
|---------------|---------------------------------------|-------------------------------------------|----------------------------------------------------------|----------|----------------------------------|---------------------------------------------------------------------|
| l4-media-host | make -C services/media configure-host | media_host_allow + MEDIA<br>HOST media-vm | /etc/media/secrets/cifs.cred installed<br>root:root 0600 | 10 min   | make -C services/media preflight | Unmount the shares; packages and directories are harmless to leave. |

### 12.1.1.6 L5 — Applications and their configuration

| Step             | Command                                                | Gate                                                                                                     | Inputs                                                                                                      | Duration       | Verify                                                                          | Rollback                                         |
|------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------|--------------------------------------------------|
| l5-vm300-apps    | make -C platform/proxmox homelab-service SERVICE=<app> | homelab_allow_service_deploy + DEPLOY<br><app> ON homelab-services                                       | /etc/homelab/secret-store/* for the app; one app per run                                                    | 10 min per app | Homepage card for the app turns green; uptime_kuma_monitors.py reports no drift | Re-deploy the app from the previous commit.      |
| l5-npm-routes    | make -C platform/proxmox homelab-service SERVICE=npm   | homelab_services_allow_npm_route_ownership + REPLACE NPM UI PROXY HOSTS WITH DECLARATIVE HOME LAB ROUTES | wildcard certificate DNS-01 credential                                                                      | 15 min         | every lab.yml proxied site answers over HTTPS                                   | Restore NPM data from the L6 application backup. |
| l5-dns-records   | manual                                                 | manual: Synology DNS has no provider                                                                     | lab.yml<br>dns_records                                                                                      | 20 min         | dig @192.168.0.20 .lab.example.com returns 192.168.0.30                         | Revert the record in DSM.                        |
| l5-observability | make -C observability deploy                           | observability_allow_deploy + DEPLOY OBSERVABILITY STACK ON observability                                 | /etc/observability/secret-store/* (L7 table)                                                                | 20 min         | make -C observability validate                                                  | Restore the observability archive (L6).          |
| l5-netbox        | make -C platform/proxmox netbox-deploy                 | netbox_allow_deploy + DEPLOY NETBOX ON netbox                                                            | NetBox secrets; on a first deployment also netbox_app_allow_first_deployment with "DEPLOY NETBOX ON VM 330" | 20 min         | make -C platform/proxmox netbox-preflight                                       | Restore the pg_dump backup (L6).                 |
| l5-netbox-seed   | manual                                                 | manual: Automation never writes to NetBox                                                                | lab.yml                                                                                                     | 15 min         | NetBox lists every lab.yml node, guest and address                              | Delete the imported objects in NetBox.           |

| Step                | Command                                         | Gate                                                                       | Inputs                                                                                                                 | Duration | Verify                                                                                                    | Rollback                                                               |
|---------------------|-------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| l5-openbao-install  | make -C security/secrets-openbao install-server | al-low_install_openbao + INSTALL OPENBAO 2.6.1 WITHOUT INITIALIZING        | signed Open-Bao 2.6.1 binary                                                                                           | 15 min   | make -C security/secrets-openbao validate-uninitialized                                                   | Stop and disable the service.                                          |
| l5-openbao-tls      | make -C security/secrets-openbao configure-tls  | al-low_configure_openbao_tls + DEPLOY OPENBAO TLS TO THREE NODES           | operator-issued node certificates from the off-line root                                                               | 15 min   | make -C security/secrets-openbao validate-uninitialized                                                   | Re-deploy the previous certificates.                                   |
| l5-openbao-ha       | make -C security/secrets-openbao configure-ha   | al-low_configure_openbao_ha + CONFIGURE OPENBAO RAFT AND VIP UNINITIALIZED | VIP 192.168.0.40 reserved                                                                                              | 15 min   | make -C security/secrets-openbao validate-uninitialized                                                   | Stop Keepalived and HAProxy.                                           |
| l5-media-arr        | make -C services/media deploy-arr               | arr_stack_allow + MEDIA DEPLOY media-vm                                    | /etc/media/secrets/*.env installed<br>root:root 0600                                                                   | 15 min   | Radarr and Prowlarr answer on workstation's loop-back forwards; up{host="media-vm"} appears in Prometheus | docker compose down in /opt/lux-media, then restore *arr backups (L6). |
| l5-media-arr-config | make -C services/media arr-plan                 | manual: Apply is manual after the plan is reviewed; no apply target exists | backend.hcl;<br>TF_VAR_radarr_api_key;<br>TF_VAR_prowlarr_api_key;<br>TF_VAR_qbittorrent_*;<br>TF_VAR_state_passphrase | 10 min   | The plan shows only the root folder, qBittorrent client, remote path mapping and Prowlarr link.           | Restore the Radarr/Prowlarr backup zip (L6).                           |
| l5-media-profiles   | make -C services/media sync-quality-profiles    | arr_stack_allow_recyclarr_sync + RECYCLARR SYNC media-vm                   | /etc/media/secrets/recyclarr.env                                                                                       | 5 min    | The preview (no gate) shows no changes after a real sync.                                                 | Restore the Radarr backup zip (L6).                                    |
| l5-plex             | manual                                          | manual: Claim, client firewall rule and library apply are attended         | Plex claim session;<br>PLEX_TOKEN;<br>first client IP                                                                  | 30 min   | python services/media/scripts/plex_libraries.py                                                           | Remove the library in Plex; NAS media is untouched.                    |

| Step           | Command | Gate                                             | Inputs                                                    | Duration | Verify                                                                  | Rollback                      |
|----------------|---------|--------------------------------------------------|-----------------------------------------------------------|----------|-------------------------------------------------------------------------|-------------------------------|
| l5-uptime-kuma | manual  | manual: Uptime Kuma 2 has no supported write API | services/homelab-services/config/uptime-kuma/monitors.yml | 20 min   | python services/homelab-services/scripts/uptime_kuma_monitors.py - dump | Remove the monitor in the UI. |

Operator actions:

- `l5-dns-records`: Create the A records in lab.yml dns\_records in Synology DNS Server.
- `l5-netbox-seed`: python services/netbox/scripts/netbox\_seed\_csv.py -out , then bulk import the three files in order.
- `l5-plex`: Claim Plex in the operator's browser, add the client-scoped firewall rule by hand, then run plex\_libraries.py plan and, with PLEX\_ALLOW\_APPLY=true and PLEX\_CONFIRMATION="PLEX LIBRARIES workstation", -apply.
- `l5-uptime-kuma`: Create each monitor in monitors.yml in the Uptime Kuma UI.

#### 12.1.1.7 L6 — Data restore (isolated first)

| Step              | Command                                        | Gate                                                                  | Inputs                                                                            | Duration                 | Verify                                                                              | Rollback                                                                             |
|-------------------|------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| l6-restore-drills | manual                                         | manual: Restores run only against an isolated VM or network, attended | docs/reference/backup-matrix.yml                                                  | 1-4 h                    | Each backup-matrix row's verify command passes on the isolated copy.                | Discard the isolated copy; production was never touched.                             |
| l6-backup-pull    | make -C platform/proxmox configure-backup-pull | allow_backup_pull + INSTALL BACKUP_PULL ON <inventory-node> <address> | dedicated public key; least-loaded mounted node; verified host pins; D free space | 10 min plus archive copy | Install the local task; size and SHA-256 match NAS before and after the first pull. | Disable only this task and separately remove only its public key; retain every copy. |
| l6-media-backups  | make -C services/media configure-backup        | media_backup_allow + MEDIA BACKUP media-vm                            | media-backups share mounted                                                       | 5 min                    | systemctl list-timers arr-backup.timer on the guest                                 | systemctl disable -now arr-backup.timer (copies are kept).                           |

Operator actions:

- `l6-restore-drills`: Follow every row of docs/reference/backup-matrix.yml, one at a time.
- `l6-backup-pull`: Follow platform/proxmox/docs/backup-pull.md; stop for the exact node confirmation.

**12.1.1.8 L7 — Attended secrets and ceremonies**

| Step               | Command | Gate                                                                      | Inputs                               | Duration | Verify                                                           | Rollback                                                              |
|--------------------|---------|---------------------------------------------------------------------------|--------------------------------------|----------|------------------------------------------------------------------|-----------------------------------------------------------------------|
| l7-openbao-init    | manual  | manual: bao operator init/unseal is an attended ceremony; never automated | five PGP recipients; threshold three | 1 h      | make -C security/secrets-openbao validate-post-init              | None automated; see security/secrets-openbao/docs/restore-runbook.md. |
| l7-operator-inputs | manual  | manual: Secrets are installed by the operator only                        | docs/reference/operator-inputs.md    | 1 h      | Each stage's secret-file metadata check passes (root:root 0600). | Rotate the affected credential.                                       |

Operator actions:

- `l7-openbao-init`: Run the ceremony by hand, then `make -C security/secrets-openbao configure-post-init` with its gate.
- `l7-operator-inputs`: Install or rotate each row of `docs/reference/operator-inputs.md`.

**12.1.1.9 L8 — Validate and enable onboot**

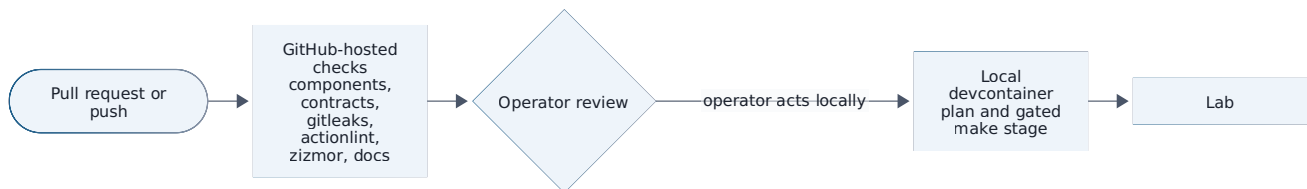
| Step                      | Command                                                      | Gate                                                            | Inputs                                                           | Duration | Verify                                                                           | Rollback                                        |
|---------------------------|--------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------|----------|----------------------------------------------------------------------------------|-------------------------------------------------|
| l8-validate-cluster       | <code>make -C platform/proxmox validate</code>               | read-only: Validation only                                      | —                                                                | 10 min   | The report shows every check passing.                                            | Not applicable.                                 |
| l8-validate-observability | <code>make -C observability validate</code>                  | read-only: Validation only                                      | —                                                                | 5 min    | <code>make -C observability validate</code> passes with <code>changed=0</code> . | Not applicable.                                 |
| l8-onboot                 | <code>make -C platform/proxmox rebuild-l1-host-config</code> | <code>pve_host_config_allow + L1 CONFIGURE pve pve1 pve2</code> | lab.yml onboot set to true only for guests whose L6 drill passed | 5 min    | A controlled node reboot brings the chosen guests back.                          | Set onboot back to false in lab.yml and re-run. |

Source: [docs/runbooks/rebuild-from-zero.md](#)

## 13 CI checks

### 13.1 Pipeline

GitHub runs offline checks on hosted runners with no lab credentials. Plans and deployments run locally from the devcontainer, one gated stage at a time; there is no GitHub runner inside the lab (ADR-0009).



**Figure 15.** GitHub checks end at review; the operator runs gated local stages

Diagram source: [docs/walkthrough/diagrams/15-cicd-pipeline.mmd](#)

Excerpt from `.github/workflows/ci.yml`, lines 1-15: not published in this copy.

`permissions: {}` at the top means every job must ask for exactly the scopes it uses. The `on:` key is quoted so YAML 1.1 parsers do not read it as `true`. Every action is pinned by full commit SHA, which a contract test enforces.

### 13.2 Local operations

The operator reviews the change, then runs its existing local Make target with the exact gate. GitHub supplies neither approvals nor credentials.

### 13.3 CI checks

GitHub checks source on hosted runners; it has no path into the lab. See [ADR-0009](#).

#### 13.3.1 Checks and permissions

`ci.yml` selects changed components and calls `_component-checks.yml`. Contracts, gitleaks and docs always run. Music Library keeps its own lint, test and import job. Actions are pinned by full SHA; workflow permissions default to empty and jobs request read access only.

Actionlint checks syntax. Zizmor is installed at a pinned version with a pinned SHA-256 and runs offline audits. There are no configured GitHub secrets, environments, lab runners, deploy dispatches or plan jobs.

#### 13.3.2 Repository setup

Private origin: [HomeLab-Private](#). Public origin: [HomeLab-Public](#). Enable Actions on the private repository using hosted runners only. Do not register a self-hosted runner or configure lab credentials. Renovate may open dependency PRs; auto-merge stays disabled. Public workflows remain withheld.

### 13.3.3 Local operations and rollback

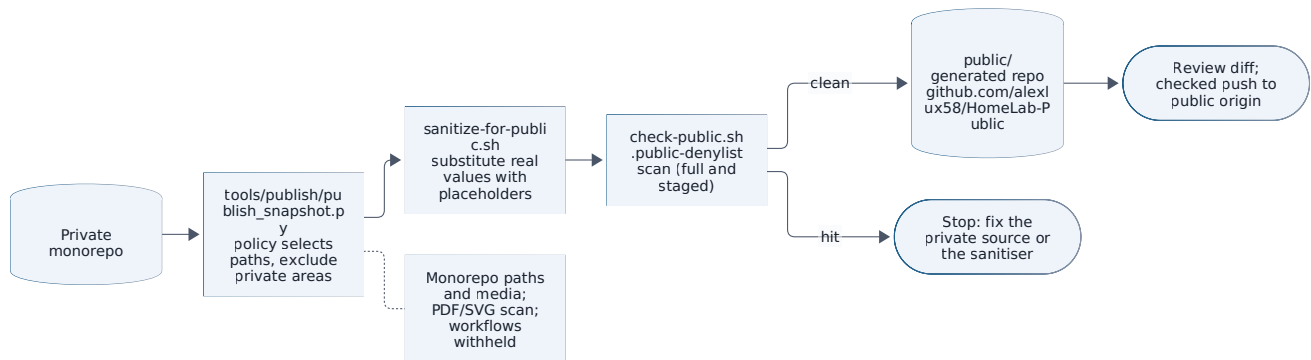
After checks pass, review the change and run one gated local Make target in the devcontainer. Approval flags and exact confirmations are the operator's. State plans and migrations follow ADR-0003 and its migration runbook. No GitHub event can authorize or launch a lab stage.

A failing check stops publication or pushing until repaired. Revert a faulty source change with a normal commit; never force-push. CI does not change production. Deployment rollback follows the component runbook.

Source: [docs/runbooks/ci-cd.md](https://docs.runbooks/ci-cd.md)

## 13.4 Publishing and documentation

A sanitised public copy is produced by `tools/publish/` (not published) (Phase 6). Its origin is the approved public repository. This document is built by `tools/docs-build/build.py` from Markdown, Mermaid sources and the generated reference. The build fails on any warning, broken link or anchor, or changed snippet source.



**Figure 16.** From private source to the sanitised public copy and this PDF

Diagram source: [docs/walkthrough/diagrams/16-publish-pipeline.mmd](https://docs.walkthrough/diagrams/16-publish-pipeline.mmd)

## 14 On-demand music library

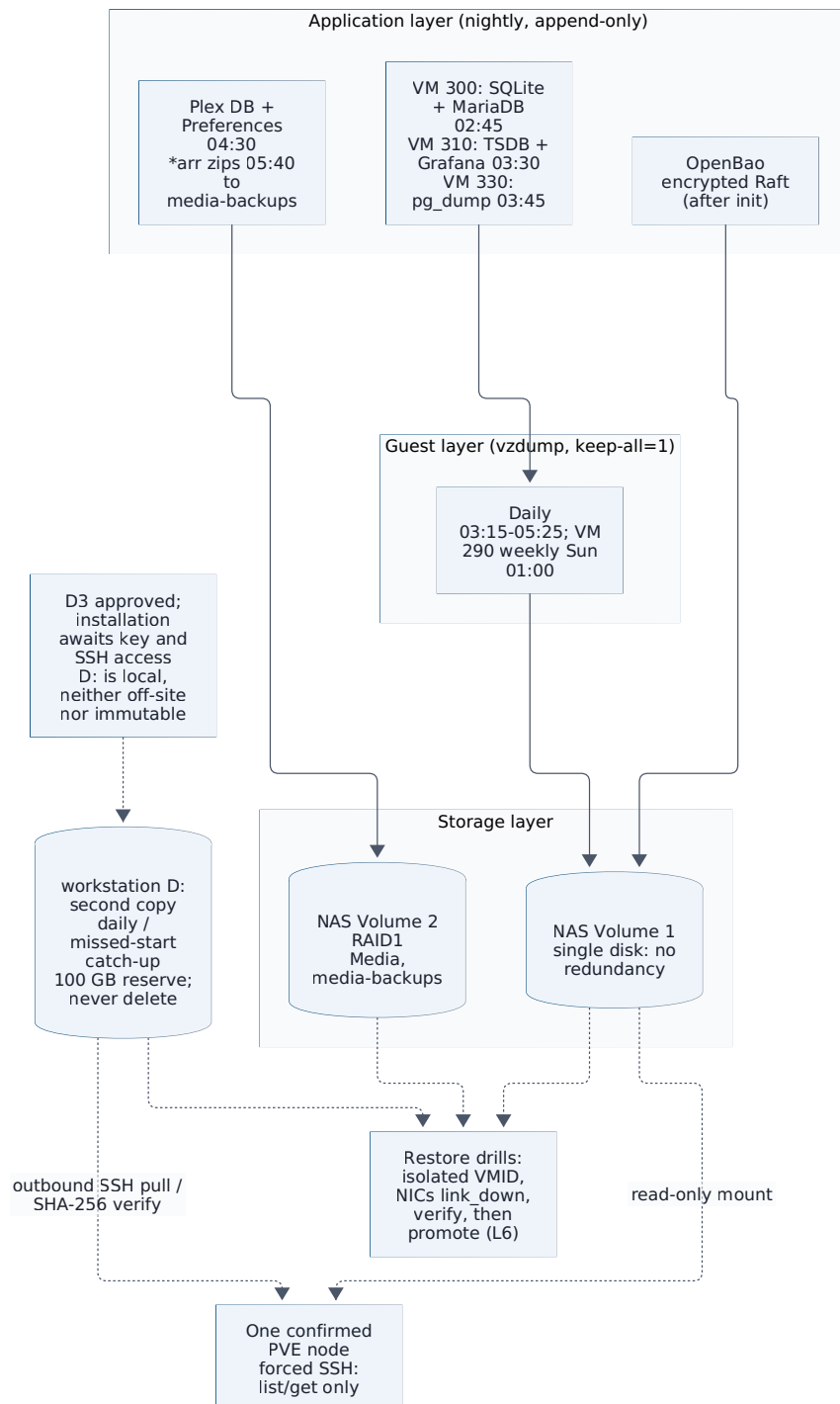
---

The private Music Library app (not published) runs on demand on workstation. Completed downloads are checksum-verified into the user's Music folder and NAS Media/Music. Tempo is estimated locally and used for ten-BPM range folders; unavailable analysis uses Unknown BPM. A sortable BPM column and editable metadata expose the result. Tempo estimation can report half/double tempo, so the operator can correct it.

The prior packaged app saved downloads inside its bundle instead of the user Music folder. Three existing tracks now have verified copies in both destinations. The desktop launcher targets the new BPM bundle; close the old app, restart, and use Organize by BPM once to redirect its existing catalog to those copies. Existing song originals and the only live catalog remain preserved. This app's source is withheld from the public reading copy.

# 15 Backup and restore

Every stateful item has a source, a destination, a schedule, a restore method and a verify step. Restores always go to an isolated guest first: a spare protected VMID with every NIC link-down. Nothing prunes. Destinations are append-only, and Proxmox storage is keep-all=1.



**Figure 17.** NAS backups, the approved read-only D: pull copy, and isolated restore paths

Diagram source: <docs/walkthrough/diagrams/14-backup-restore.mmd>

The **D3 pull stage** restricts one key to list/get on the NAS-mounted archive directory of one confirmed node. workstation initiates SSH and verifies hashes before publishing each D: copy; its daily task catches missed starts while awake, never wakes the PC and stops below a 100 GB reserve. The exact pve1 installation confirmation was supplied on 2026-09-30, but the public key and permitted SSH execution remain prerequisites; neither installation nor a verified D: archive is claimed. Copies are never pruned. D: is a second on-site copy exposed to desktop ransomware. Future off-site data transfer must be an outbound push; GitHub has no path into the lab (ADR-0009).

### **DANGER — DESTRUCTIVE STAGE**

Never delete or prune a backup archive, and never restore over a live VMID. The two archives of VM 297 ( `ubuntu-vm`, destroyed on or before 2026-09-19) are the only remaining copy of that guest and must be kept.

## 15.1 Backup and restore matrix

Rendered from [docs/reference/backup-matrix.yml](#); edit that file, then run `make docs-reference`.

| Item                | What                                                                                                                        | Schedule                                                                                   | Destination                                                                                      | Restore                                                                                                                                                                                                                                                                                      | Verify                                                                                                                                     | Status               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| proxmox-guests      | Whole-guest vmdump archives for VMs 290, 300, 310, 320-322, 330                                                             | inventory/ lab.yml<br>backup_jobs<br>(daily; VM 290 weekly<br>Sunday 01:00)                | synology-backup NFS,<br>NAS Volume 1<br>(single disk —<br>no redundancy)                         | qmrestore -<br>storage -<br>unique 1,<br>every NIC<br>link_down=1,<br>then inspect                                                                                                                                                                                                           | zstdcat   vma<br>verify - ; plus<br>the archive's<br>SHA-256 side-<br>car                                                                  | backup-re-<br>stored |
| proxmox-second-copy | Newest finalized vmdump archive per VMID, retained on workstation                                                           | Daily 06:30<br>while logged<br>in; catch up<br>after missed<br>start; never<br>wake the PC | D:/homelab-backups on<br>workstation<br>(second on-<br>site copy)                                | Isolated qm-<br>restore from<br>the retained D<br>copy; plat-<br>form/proxmox/<br>docs/backup-<br>pull.md                                                                                                                                                                                    | Client size and<br>SHA-256<br>match NAS list<br>before and<br>after;<br>NAS_UN-<br>CHANGED=true                                            | declared             |
| vm300-app-data      | NPM, Linking,<br>Speedtest<br>Tracker SQLite; Uptime<br>Kuma 2 Mariadb dump;<br>Homepage/<br>NPM config<br>and certificates | daily 02:45<br>(homelab-app-<br>backup.timer)                                              | /srv/homelab-backups/staging on VM<br>300, carried<br>off-guest by<br>the 03:15 VM<br>300 vmdump | On an isolated<br>restore of VM<br>300: copy<br>each SQLite<br>file back under /srv/<br>homelab/ with<br>the app<br>stopped; for<br>Uptime Kuma,<br>gunzip -c uptime-kuma-all-<br>data-bases.sql.gz  <br>docker exec -i<br>uptime-kuma<br>mariadb -<br>socket=/app/<br>data/run/mariadb.sock | sqlite3<br>'PRAGMA integrity_check'<br>returns ok;<br>gunzip -t the<br>Kuma dump;<br>uptime_kuma_monitors.py -<br>dump reports<br>no drift | backup-re-<br>stored |

| Item                | What                                                                             | Schedule                                                                                 | Destination                                                                         | Restore                                                                                                                               | Verify                                                                                                | Status          |
|---------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------|
| vm310-observability | Prometheus TSDB snapshot, grafana.db, configuration                              | daily 03:30 (homelab-observability-backup.timer)                                         | /srv/observability/backups on VM 310, carried off-guest by the 04:15 VM 310 vzdump  | make -C observability restore-drill ARCHIVE=/srv/observability/backups/ (never overwrites live data)                                  | The restore drill's checks pass; homelab_observability_backup_last_success_timestamp_seconds is fresh | backup-restored |
| vm330-netbox        | NetBox PostgreSQL (plain SQL, serializable-deferrable) and media/reports/scripts | daily 03:45 (netbox-app-backup.timer)                                                    | /srv/netbox-backups/staging on VM 330, carried off-guest by the 04:45 VM 330 vzdump | On an isolated VM 330 copy with an empty database: gunzip -c /netbox.sql.gz   docker exec -i netbox-postgres psql -U netbox -d netbox | gunzip -t the dump; NetBox login works and object counts match the source                             | backup-restored |
| openbao-raft        | Encrypted OpenBao Raft snapshots                                                 | configured only after initialization (make -C security/secrets-openbao configure-backup) | Synology, encrypted with an age identity kept outside OpenBao and the NAS           | security/secrets-openbao/docs/restore-runbook.md in a protected VM on an isolated bridge                                              | make -C security/secrets-openbao inspect-backup SNAPSHOT=AGE_IDENTITY=                                | manual          |
| arr-apps            | Radarr and Prowlarr scheduled backup zips (database and config)                  | daily 05:40 (arr-backup.timer); only while workstation is awake                          | NAS media-backups share (Volume 2) under arr//, with SHA-256 sidecars               | On an isolated guest, Radarr/Prowlarr System > Backup > Restore from the copied zip                                                   | sha256sum -c .sha256; python3 -m zipfile -t                                                           | backup-restored |
| plex-metadata       | Plex database backups and Preferences.xml (holds the Plex token)                 | daily 04:30 while the desktop user is logged on (plex role, gated)                       | NAS media-backups share under plex/ (operator-only ACL)                             | Stop Plex; copy the dated com.plexapp.plugins.library.db backup and Preferences.xml back; start Plex                                  | python services/media/scripts/plex_libraries.py check, then plan shows no drift                       | backup-restored |

| Item              | What                                                        | Schedule                                                                      | Destination                                                                 | Restore                                                                                                                                  | Verify                                                                                            | Status   |
|-------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------|
| media-library     | Movies and Downloads on the NAS Media share                 | not applicable                                                                | none — protected by the RAID1 mirror only                                   | Not recreated from backup: rebuild the mirror from the healthy disk after replacing bay 3 (platform/nas/docs/synology-storage-health.md) | DSM Storage Manager shows the pool healthy; no Btrfs snapshots exist, and Disk 3 has failed SMART | manual   |
| qbittorrent-state | qBittorrent settings and torrent list on Windows            | not applicable                                                                | none                                                                        | Reinstall and set the Web UI port, credentials and M:path by hand (L0/L7)                                                                | Radarr's download-client test succeeds                                                            | manual   |
| media-guest-disk  | media-vm system and data disks                              | not applicable                                                                | none — rebuilt, not restored                                                | Rebuild with L0 (cloud-init template) + L4/L5 (Ansible) and restore arr-apps above                                                       | make -C services/media preflight, then the L5 media verifications                                 | declared |
| tofu-state        | Encrypted OpenTofu state for every root                     | Operator copy after every verified state write; unique names retained forever | Encrypted timestamped copies on NAS proxmox-cluster-backups/opentofu-state/ | Restore an encrypted copy to D:/homelab-state; otherwise re-adopt by import                                                              | SHA-256 matches both copies; strict encryption plan shows no changes                              | manual   |
| nas-configuration | DSM configuration (shares, accounts, DNS zone, NFS exports) | after every DSM change                                                        | operator's encrypted storage (manual DSM configuration export)              | DSM Control Panel > Update & Restore > Configuration Backup                                                                              | Shares, accounts and DNS records in lab.yml exist after restore                                   | manual   |

Source: <docs/reference/generated/backups.md>

# 16 Troubleshooting

| Symptom                                                      | Likely cause                                                | First response                                                                                |
|--------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| A stage fails immediately with “Set ... to the exact string” | Gate not passed, or the confirmation differs by a character | Copy the expected string from <a href="#">Appendix C</a> ; pass it in <code>EXTRA_JSON</code> |
| Approval set but destructive tasks skipped                   | <code>--tags all,destructive</code> missing                 | Use the Make target, which passes the tags                                                    |
| SSH fails with a changed host key                            | Cloud-init re-ran on a stopped guest, or a reinstall        | Verify through <code>qm guest exec ... ssh-keygen -lf</code> ; the operator re-pins           |
| Private names fail on a guest, public names work             | NAS DNS down, or the routing-domain drop-in missing         | Check <code>resolvectl status</code> ; never point the guest wholesale at the NAS             |
| A freshness alert never fires                                | Its input series does not exist                             | Query the metric first; check <code>node_textfile_scrape_error</code>                         |
| Backup timer failed for days, no alert                       | Unit failure is not a metric                                | Check <code>journalctl -u &lt;timer&gt;</code> ; the freshness metric covers it now           |
| <code>make docs</code> fails with “snippet ... changed”      | The source lines moved or changed                           | Review the new lines, fix the range, run <code>build.py --fill</code>                         |
| CI <code>inventory contract</code> fails                     | A live identity changed in one place only                   | Change <code>inventory/lab.yml</code> and the consumer together                               |

The facts behind most of these rows were learned against the live hosts. They are recorded once, in the lessons-learned reference, included here:

## 16.1 Lessons learned against the live lab

Verified against the live hosts. Do not re-derive these. Moved verbatim from the root `CLAUDE.md` in Phase 2 (2026-09-29).

### 16.1.1 Environment facts learned the hard way

Verified against the live hosts. Do not re-derive these.

- **A metric can be written and still never exist.** `backup.sh` writes `homelab_observability_backup_last_success_timestamp_seconds` into `/var/lib/node_exporter/textfile/` as mode **0600 root:root**. The `node_exporter` container runs as **nobody** and cannot read it, so `node_textfile_scrape_error` = 1 and Prometheus has **no such series**. Discovered 2026-09-19. Consequence: any alert written against that metric is **vacuous** — it evaluates against nothing and can never fire, which looks identical to “healthy”. Whenever you add an alert, first confirm its input series actually returns data, and alert on `node_textfile_scrape_error > 0` so the collector cannot fail silently again.
- **A backup can fail for days while monitoring is perfectly healthy.** VM 310’s application backup failed six consecutive nights, 2026-09-11 → 09-16, with `curl: (7) Failed to connect to 192.168.0.31 port 9090`. Prometheus had **continuous data with no gap over 10 minutes** throughout, so the stack was not down and the “stack was down” explanation is wrong. **Root cause remains unresolved.** This is the second unexplained multi-day backup gap, after 2026-08-28 → 09-05. `systemctl --failed` is useless retrospectively: later successful runs clear the failed state.

- `qm set <vmid> --nameserver ...` on a **STOPPED** cloud-init guest regenerates the cloud-init drive, and on the next boot cloud-init re-runs and **REGENERATES THE GUEST'S SSH HOST KEYS**. This happened on 2026-09-06 to VMs 320 `bao-1` and 321 `bao-2`: their pinned `known_hosts` fingerprints were invalidated by a routine nameserver correction. Guests that were **running** when the same change was applied (322, 330) kept their keys, because cloud-init never re-ran. Before changing cloud-init settings on a stopped guest, expect a host key change and plan to re-verify the fingerprint out of band.
- **A guest's SSH host key can be verified out of band through the hypervisor** with `qm guest exec <vmid> -- /usr/bin/ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub`. This does not traverse the network SSH path, so it is a valid independent channel for confirming that a changed key is the guest's own key and not an interception. It is the same method used to verify VM 300 in Stage C3. It still does **not** authorise auto-accepting the key — the operator updates the pin.
- `pvesm status|list --output-format json` **does not exist** on PVE 8.2 or 9.1. Use `pvesh get /nodes/<node>/storage ...` instead.
- `pvesm status` **column 6 is Available**, not `$5-$6`.
- `/etc/pve/qemu-server, lxc and local` are **symlinks** into `/etc/pve/nodes/<node>/`. `tar` needs `-h` or the guest configs come out empty while the play still reports success.
- **A regex backreference inside a folded (>-) or literal (|-) YAML scalar is passed through verbatim** by Ansible — `regex_search(..., '\1')` yields the string `\1`. Plain and double-quoted scalars are fine. This silently broke the pve8to9 failure counter, which then evaluated to 0 and would have approved a failed upgrade. A test now guards it.
- `mount.nfs` **rejects vers= and nfsvers= together**. The version is negotiated by the probe mount, not hardcoded.
- `ansible.builtin.unarchive` **requires GNU tar**, absent on a stock macOS controller. Use `tar -xzf` directly.
- `community.general.yaml` **stdout callback was removed** in v12; use the built-in default callback with `result_format = yaml`.
- **Proxmox CLI tools have no --help**. Use `pvesh usage <api-path> -v`.
- **Ansible's never tag**: `--tags all,destructive` runs everything including never-tagged destructive tasks. Verified.
- The backup role must create `/root/.pve-migration/manifests` before the first archive. The first live Stage 3 run exposed this omission after writing and hashing EVENG; the archive was kept and the role now creates the directory.
- Bash `PIPESTATUS` cannot be read in the parent after wrapping a pipeline in command substitution. The first archive-verification run exposed this. The verifier now runs the pipeline in the current shell and captures both statuses.
- `gathering=smart` can reuse a cached `ansible_date_time` in evidence. Gathering is now explicit, and backup timestamps come from the archive file's own mtime.
- Ansible's key/value `-e` parser splits multi-word values even when the shell delivered them as one argument. Approval examples now use `EXTRA_JSON` so exact confirmation phrases are passed as structured JSON.
- Role defaults are scoped to the play that invokes the role. The first full Stage 4b run verified all four archives but its localhost adjudication could not see `backup_verify_generations`; the variable now lives in `group_vars/all`.

### 16.1.2 Working style that has paid off here

Measure before asserting. Two plausible root-cause hypotheses (snapshots, then disk failure) were both wrong and were only caught by running commands against the live systems. Prefer a read-only command over an inference, and say plainly when a previous claim turns out to be wrong.

`du` on the NAS is close to useless — ARM CPU, millions of files, scans run for hours. Prefer DSM's own reporting, `btrfs` tooling, or per-child parallel scans.

Source: [docs/reference/lessons-learned.md](#)

# 17 Glossary

| Term                    | Meaning                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------|
| Allow flag              | A boolean such as <code>netbox_allow_deploy</code> ; defaults <code>false</code>                  |
| Confirmation            | An exact string, often embedding the host, that must match before a stage runs                    |
| EXTRA_JSON              | The Make variable that carries approvals as JSON to <code>ansible-playbook -e</code>              |
| Import-first            | An OpenTofu root that adopts existing guests with <code>import</code> blocks before managing them |
| Rebuild-only root       | A create-mode OpenTofu root used only after total loss ( <a href="#">ADR-0007</a> )               |
| Rung                    | One level (L0–L8) of the rebuild ladder                                                           |
| Routing domain          | A systemd-resolved <code>~domain</code> that sends only that zone to a given server               |
| <code>keep-all=1</code> | Proxmox prune policy that retains every archive                                                   |
| Isolated restore        | A restore into a spare protected VMID with every NIC link-down                                    |
| vzdump                  | Proxmox's whole-guest backup; archives are <code>.vma.zst</code>                                  |
| VIP                     | The Keepalived virtual address (192.168.0.40) in front of OpenBao                                 |
| Ratchet                 | A list of accepted exceptions that tests only allow to shrink                                     |
| Split horizon           | The lab zone resolves only inside the LAN, from the NAS                                           |

# 18 Appendix A: Make targets

Generated from every Makefile by [tools/docs/generate\\_reference.py](#). Targets without help text print `—`.

## 18.1 Make targets

### 18.1.1 repository root

| Target         | What it does |
|----------------|--------------|
| help           | —            |
| rebuild-plan   | —            |
| docs-check     | —            |
| docs-reference | —            |
| docs-diagrams  | —            |
| docs           | —            |

### 18.1.2 observability

| Target            | What it does                                                              |
|-------------------|---------------------------------------------------------------------------|
| help              | List independently gated stages                                           |
| setup             | Install local controller tooling                                          |
| install-hooks     | Install pinned pre-commit secret and safety checks                        |
| lint              | Validate YAML, Ansible, Compose, Prometheus, and secret hygiene           |
| test              | Run offline repository tests                                              |
| syntax            | Syntax-check every playbook without contacting a host                     |
| check             | Run all offline validation                                                |
| preflight         | Read-only discovery; writes artifacts/preflight.json on controller        |
| bootstrap         | Harden the already-created observability VM; no stack deployment          |
| deploy            | Deploy central stack only after reviewing preflight and providing secrets |
| validate          | Read-only runtime validation                                              |
| update-dashboards | Gated dashboard/rule update without restarting containers                 |
| backup            | Trigger one application-aware backup on the observability VM              |
| restore-drill     | Validate a named backup archive without overwriting live data             |

### 18.1.3 platform/nas

| Target    | What it does |
|-----------|--------------|
| help      | —            |
| setup     | —            |
| lint      | —            |
| test      | —            |
| syntax    | —            |
| check     | —            |
| plan      | —            |
| preflight | —            |
| deploy    | —            |

### 18.1.4 platform/proxmox

| Target                 | What it does                                                         |
|------------------------|----------------------------------------------------------------------|
| help                   | Show this help                                                       |
| setup                  | Create the local virtualenv and install controller dependencies      |
| lint                   | Run yamllint + ansible-lint + ruff + shellcheck (offline)            |
| test                   | Run the offline unit test-suite                                      |
| syntax                 | Syntax-check every migration playbook without contacting a host      |
| check                  | Run all offline validation                                           |
| tree                   | Print the repository tree                                            |
| clean-artifacts        | Remove generated reports (NEVER touches backups on the NAS)          |
| configure-backup-pull  | Gated: authorize the restricted workstation pull key on ONE node     |
| validate-ssh           | Stage 0a: prove existing SSH key auth works on all three hosts       |
| discover               | Stage 0b: read-only discovery of all three hosts                     |
| nas-preflight          | Stage 1: validate the Synology NFS target from all three hosts       |
| capture-config         | Stage 2: capture sanitized host configuration from .11 and .12       |
| backup-initial         | Stage 3: snapshot-mode backup (usage: make backup-initial HOST=pve1) |
| backup-final           | Stage 4: stopped-guest backup (usage: make backup-final HOST=pve1)   |
| verify-backups-initial | Stage 3b: verify the initial generation before final shutdown        |
| verify-backups         | Stage 4b: checksum + vma/tar verify every archive                    |

| Target                    | What it does                                                              |
|---------------------------|---------------------------------------------------------------------------|
| prepare-pve3              | Stage 5a: emit the .13 clean-install checklist / validate after reinstall |
| test-restore              | Legacy plan only (not used by the current rebuild-later migration)        |
| precheck-upgrade-pve1     | Stage 6a: pve8to9 -full + repository change report for .11                |
| upgrade-pve1              | Stage 6b: gated PVE 8 -> 9 upgrade of .11 (requires console access)       |
| create-cluster            | Stage 7: create the cluster on .11 only                                   |
| join-pve3                 | Stage 8: join .13 (requires allow_join_pve3 + confirmation string)        |
| precheck-destroy-pve2     | Stage 9a: verify every gate before .12 may be clean-installed             |
| postinstall-pve2          | Stage 9b: validate .12 after its manual clean install                     |
| join-pve2                 | Stage 9c: join .12 to the cluster                                         |
| build-restore-plan        | Legacy plan only: generate the retired .12 restore plan                   |
| restore-pve2              | Legacy plan only: restore .12 guests (not in current migration)           |
| start-restored-guests     | Legacy plan only: start restored .12 guests                               |
| lan-scan                  | Read-only LAN sweep -> artifacts/lan-inventory.json                       |
| lan-site                  | Build the static LAN inventory page -> site/index.html                    |
| lan-preview               | Serve site/ locally at http://127.0.0.1:8899                              |
| lan-deploy                | Copy the page to the Synology (requires Web Station + a Volume 1 docroot) |
| homelab-preflight         | Read-only validation of VM 300 before reconciliation                      |
| homelab-base              | Reconcile VM 300 Docker, hardening, and filesystem prerequisites          |
| homelab-service           | Reconcile one app (usage: make homelab-service SERVICE=homepage)          |
| homelab-backups           | Reconcile VM 300 application-aware backup timer                           |
| netbox-preflight          | Read-only validation of the NetBox guest before reconciliation            |
| netbox-base               | Reconcile NetBox guest Docker, hardening, and prerequisites               |
| netbox-deploy             | Deploy NetBox (requires the documented approval, see docs/netbox-plan.md) |
| netbox-backups            | Reconcile the NetBox application-aware backup timer                       |
| netbox-terraform-fmt      | Check planned NetBox guest Terraform formatting                           |
| netbox-terraform-init     | Initialize the pinned Proxmox provider without a backend                  |
| netbox-terraform-validate | Validate the planned NetBox guest Terraform configuration                 |
| netbox-terraform-plan     | Plan the planned NetBox guest. Apply stays a manual, reviewed step.       |

| Target                  | What it does                                                                   |
|-------------------------|--------------------------------------------------------------------------------|
| terraform-fmt           | Check Home Lab guest Terraform formatting                                      |
| terraform-init          | Initialize the pinned Proxmox provider without a backend                       |
| terraform-validate      | Validate Home Lab guest Terraform configuration                                |
| terraform-plan          | Import-first, non-destructive Terraform plan for VMs 300 and 310               |
| rebuild-l0-answer-files | L0: render Proxmox answer files (dry run unless WRITE=1)                       |
| rebuild-l1-host-config  | L1: vzdump jobs, onboot, startup from lab.yml (gated by EXTRA_JSON)            |
| rebuild-guests-fmt      | Check rebuild-only root formatting                                             |
| rebuild-guests-validate | Validate the rebuild-only root (init -backend=false)                           |
| rebuild-guests-test     | Offline tofu test with a mocked provider                                       |
| rebuild-guests-plan     | L3 operator plan; needs REBUILD_VARS='-var-file=...' with names + confirmation |
| validate                | Stage 11: full three-node cluster validation + migration report                |

### 18.1.5 security/aaa-freeradius

| Target    | What it does |
|-----------|--------------|
| help      | —            |
| setup     | —            |
| lint      | —            |
| test      | —            |
| syntax    | —            |
| check     | —            |
| plan      | —            |
| preflight | —            |
| deploy    | —            |

### 18.1.6 security/secrets-openbao

| Target         | What it does                                            |
|----------------|---------------------------------------------------------|
| help           | Show independently gated targets                        |
| setup          | Install pinned offline-controller dependencies          |
| lint           | Run repository, Python, YAML, and Ansible safety checks |
| test           | Run offline unit tests                                  |
| syntax         | Syntax-check every playbook without contacting a host   |
| check          | Run all offline validation                              |
| terraform-fmt  | Check Terraform formatting                              |
| terraform-init | Initialize the pinned provider without a backend        |

| Target                 | What it does                                                                 |
|------------------------|------------------------------------------------------------------------------|
| terraform-validate     | Validate Terraform without contacting Proxmox                                |
| terraform-plan         | Produce a reviewed plan; no apply target exists                              |
| discover-plan          | Refresh sanitized secret-name and Git-history inventories                    |
| preflight              | Read-only guest validation after first boot                                  |
| prepare-hosts          | Gated Ubuntu hardening and firewall stage                                    |
| install-server         | Gated signed OpenBao binary and systemd installation                         |
| configure-tls          | Gated operator-supplied TLS deployment                                       |
| configure-ha           | Gated HAProxy/Keepalived and OpenBao configuration                           |
| validate-uninitialized | Prove TLS works and OpenBao remains uninitialized                            |
| configure-post-init    | Gated policies/auth/engines after the attended ceremony                      |
| validate-post-init     | Read-only proof of quorum, mounts, auth, audit, and TLS                      |
| configure-backup       | Gated encrypted Raft snapshot timer installation                             |
| configure-monitoring   | Gated OpenBao telemetry and audit shipping integration                       |
| backup-now             | Trigger one encrypted snapshot on the elected backup node                    |
| inspect-backup         | Offline decrypt/inspect of one snapshot (SNAPSHOT and AGE_IDENTITY required) |
| recovery-bundle        | Build a secret-free configuration recovery bundle                            |

### 18.1.7 `services/homelab-services`

| Target    | What it does |
|-----------|--------------|
| help      | —            |
| setup     | —            |
| lint      | —            |
| test      | —            |
| syntax    | —            |
| check     | —            |
| plan      | —            |
| preflight | —            |
| deploy    | —            |
| backup    | —            |

### 18.1.8 `services/media`

| Target | What it does                                               |
|--------|------------------------------------------------------------|
| help   | List the media stages                                      |
| setup  | Create the venv and install pinned tooling and collections |
| lint   | yamllint, ansible-lint, ruff                               |

| Target                | What it does                                                          |
|-----------------------|-----------------------------------------------------------------------|
| test                  | Offline pytest                                                        |
| syntax                | Syntax-check every playbook without contacting a host                 |
| check                 | Run all offline validation                                            |
| plan                  | Offline: validate and test the *arr root with mocked providers        |
| preflight             | Read-only media guest preflight                                       |
| configure-host        | Gated: packages, directories, NAS automounts                          |
| deploy-arr            | Gated: deploy the digest-pinned *arr Compose project                  |
| deploy                | Alias of deploy-arr (one stage)                                       |
| sync-quality-profiles | Recyclarr -preview, or a real sync behind its own gate                |
| check-plex            | Read-only Plex check; backup task behind its gate                     |
| configure-backup      | Gated: nightly append-only *arr backup copy                           |
| arr-fmt               | Check *arr root formatting                                            |
| arr-validate          | Validate the *arr root (init -backend=false)                          |
| arr-test              | Offline tofu test with mocked providers                               |
| arr-plan              | Operator plan from workstation; needs backend.hcl and TF_VAR_* inputs |

### 18.1.9 services/netbox

| Target    | What it does |
|-----------|--------------|
| help      | —            |
| setup     | —            |
| lint      | —            |
| test      | —            |
| syntax    | —            |
| check     | —            |
| plan      | —            |
| preflight | —            |
| deploy    | —            |
| backup    | —            |

### 18.1.10 tests

| Target | What it does |
|--------|--------------|
| help   | —            |
| setup  | —            |
| lint   | —            |
| test   | —            |

| Target | What it does |
|--------|--------------|
| syntax | —            |
| check  | —            |

Source: <docs/reference/generated/make-targets.md>

## 19 Appendix B: Workflows

---

### 19.1 Workflows

| Workflow                           | Triggers           | Job           | Runner       | Environment |
|------------------------------------|--------------------|---------------|--------------|-------------|
| <code>_component-checks.yml</code> | workflow_call      | checks        | ubuntu-24.04 | —           |
| <code>ci.yml</code>                | push, pull_request | changes       | ubuntu-24.04 | —           |
| <code>ci.yml</code>                | push, pull_request | component     | reusable     | —           |
| <code>ci.yml</code>                | push, pull_request | contracts     | ubuntu-24.04 | —           |
| <code>ci.yml</code>                | push, pull_request | secrets       | ubuntu-24.04 | —           |
| <code>ci.yml</code>                | push, pull_request | docs          | ubuntu-24.04 | —           |
| <code>ci.yml</code>                | push, pull_request | music-library | ubuntu-24.04 | —           |

Source: [docs/reference/generated/workflows.md](https://docs.reference/generated/workflows.md)

## 20 Appendix C: Approval gates

### 20.1 Approval gates

Every gate defaults the flag to `false` and the confirmation to `""`. Expressions using `inventory_hostname` are filled in with the target host.

| Flag                                              | Confirmation variable                                    | Expected confirmation                                            | Asserted in                                                                            |
|---------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <code>observability_allow_bootstrap</code>        | <code>observability_bootstrap_confirmation</code>        | ('BOOTSTRAP' ~ inventory_hostname)                               | <code>observability/ansible/playbooks/20-bootstrap.yml</code>                          |
| <code>observability_allow_deploy</code>           | <code>observability_deploy_confirmation</code>           | ('DEPLOY OBSERVABILITY STACK ON' ~ inventory_hostname)           | <code>observability/ansible/playbooks/40-deploy.yml</code>                             |
| <code>allow_observability_dashboard_update</code> | <code>observability_dashboard_update_confirmation</code> | UPDATE HOME LAB OBSERVABILITY DASHBOARDS                         | <code>observability/ansible/playbooks/41-update-dashboards.yml</code>                  |
| <code>allow_configure_nfs_storage</code>          | –                                                        | flag only                                                        | <code>platform/nas/ansible/roles/synology_nfs/tasks/configure.yml</code>               |
| <code>allow_repo_change_pve1</code>               | <code>repo_change_confirmation_pve1</code>               | CHANGE REPOSITORIES pve1 192.168.0.11                            | <code>platform/proxmox/ansible/playbooks/migration/41-upgrade-pve1.yml</code>          |
| <code>allow_upgrade_pve1</code>                   | <code>upgrade_confirmation_pve1</code>                   | UPGRADE pve1 192.168.0.11 TO PVE 9                               | <code>platform/proxmox/ansible/playbooks/migration/41-upgrade-pve1.yml</code>          |
| <code>allow_destroy_pve2</code>                   | <code>destroy_confirmation_pve2</code>                   | DESTROY pve2 192.168.0.12 AND DISCARD ALL FIVE GUESTS            | <code>platform/proxmox/ansible/playbooks/migration/71-postinstall-pve2.yml</code>      |
| <code>allow_restore_pve2</code>                   | <code>restore_confirmation_pve2</code>                   | RESTORE pve2 GUESTS FROM VERIFIED BACKUP                         | <code>platform/proxmox/ansible/playbooks/migration/80-restore-pve2.yml</code>          |
| <code>allow_start_restored_guests</code>          | <code>start_restored_confirmation</code>                 | START RESTORED GUESTS ON pve2                                    | <code>platform/proxmox/ansible/playbooks/migration/81-start-restored-guests.yml</code> |
| <code>allow_backup_pull</code>                    | <code>backup_pull_confirmation</code>                    | 'INSTALL BACKUP PULL ON' ~ inventory_hostname ~ ' ~ ansible_host | <code>platform/proxmox/ansible/roles/backup_pull/tasks/main.yml</code>                 |
| <code>allow_create_cluster</code>                 | <code>create_cluster_confirmation</code>                 | CREATE CLUSTER homelab ON pve1 192.168.0.11                      | <code>platform/proxmox/ansible/roles/pve_cluster/tasks/main.yml</code>                 |
| <code>pve_host_config_allow</code>                | <code>pve_host_config_confirmation</code>                | L1 CONFIGURE {{ ansible_play_hosts_all   sort   join(' ') }}     | <code>platform/proxmox/ansible/roles/pve_host_config/tasks/main.yml</code>             |

| Flag                                       | Confirmation variable                     | Expected confirmation                                                      | Asserted in                                                             |
|--------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| pve_join_allow                             | pve_join_confirmation                     | set per stage by the calling playbook ( pve_join_confirmation_required )   | platform/proxmox/ansible/roles/pve_join/tasks/main.yml                  |
| allow_guest_shutdown                       | guest_shutdown_confirmation               | SHUTDOWN GUESTS FOR FINAL BACKUP                                           | platform/proxmox/ansible/roles/vz-dump_backup/tasks/shutdown_guests.yml |
| allow_prepare_openbao_hosts                | prepare_openbao_hosts_confirmation        | PREPARE OPENBAO HOSTS 320 321 322                                          | security/secrets-openbao/ansible/playbooks/10-prepare-hosts.yml         |
| allow_install_openbao                      | install_openbao_confirmation              | INSTALL OPENBAO 2.6.1 WITHOUT INITIALIZING                                 | security/secrets-openbao/ansible/playbooks/20-install-server.yml        |
| allow_configure_openbao_tls                | configure_openbao_tls_confirmation        | DEPLOY OPENBAO TLS TO THREE NODES                                          | security/secrets-openbao/ansible/playbooks/30-configure-tls.yml         |
| allow_configure_openbao_ha                 | configure_openbao_ha_confirmation         | CONFIGURE OPENBAO RAFT AND VIP UNINITIALIZED                               | security/secrets-openbao/ansible/playbooks/40-configure-ha.yml          |
| allow_configure_openbao_post_init          | configure_openbao_post_init_confirmation  | CONFIGURE OPENBAO POST INIT NO SECRET VALUES                               | security/secrets-openbao/ansible/playbooks/60-configure-post-init.yml   |
| allow_configure_openbao_backup             | configure_openbao_backup_confirmation     | CONFIGURE ENCRYPTED OPENBAO RAFT BACKUPS                                   | security/secrets-openbao/ansible/playbooks/70-configure-backup.yml      |
| allow_configure_openbao_monitoring         | configure_openbao_monitoring_confirmation | CONFIGURE OPENBAO MONITORING AND AUDIT SHIPPING                            | security/secrets-openbao/ansible/playbooks/80-configure-monitoring.yml  |
| allow_run_openbao_backup                   | run_openbao_backup_confirmation           | RUN ONE OPENBAO RAFT BACKUP                                                | security/secrets-openbao/ansible/playbooks/90-backup-now.yml            |
| homelab_allow_guest_config                 | homelab_guest_config_confirmation         | ( 'CONFIGURE GUEST' ~ inventory_hostname )                                 | services/homelab-services/ansible/playbooks/30-configure-guest.yml      |
| homelab_allow_service_deploy               | homelab_service_deploy_confirmation       | ( 'DEPLOY' ~ homelab_services_service_name ~ ' ON ' ~ inventory_hostname ) | services/homelab-services/ansible/playbooks/40-deploy-service.yml       |
| homelab_allow_backup_config                | homelab_backup_config_confirmation        | ( 'CONFIGURE BACKUPS ON' ~ inventory_hostname )                            | services/homelab-services/ansible/playbooks/50-configure-backups.yml    |
| homelab_services_allow_npm_route_ownership | homelab_services_npm_route_confirmation   | REPLACE NPM UI PROXY HOSTS WITH DECLARATIVE HOME LAB ROUTES                | services/homelab-services/ansible/roles/homelab_services/tasks/main.yml |
| arr_stack_allow                            | arr_stack_confirmation                    | ( 'MEDIA DEPLOY' ~ inventory_hostname )                                    | services/media/ansible/roles/arr_stack/tasks/main.yml                   |

| Flag                               | Confirmation variable                     | Expected confirmation                         | Asserted in                                                |
|------------------------------------|-------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
| media_backup_allow                 | media_backup_confirmation                 | ('MEDIA BACKUP' ~ inventory_hostname)         | services/media/ansible/roles/media_backup/tasks/main.yml   |
| media_host_allow                   | media_host_confirmation                   | ('MEDIA HOST' ~ inventory_hostname)           | services/media/ansible/roles/media_host/tasks/main.yml     |
| –                                  | plex_backup_confirmation                  | ('PLEX BACKUP' ~ inventory_hostname)          | services/media/ansible/roles/plex/tasks/main.yml           |
| net-box_allow_guest_config         | net-box_guest_config_confirmation         | ('CONFIGURE GUEST' ~ inventory_hostname)      | services/netbox/ansible/playbooks/31-configure-guest.yml   |
| netbox_allow_deploy                | net-box_deploy_confirmation               | ('DEPLOY NETBOX ON' ~ inventory_hostname)     | services/netbox/ansible/playbooks/40-deploy.yml            |
| net-box_allow_backup_config        | net-box_backup_config_confirmation        | ('CONFIGURE BACKUPS ON' ~ inventory_hostname) | services/netbox/ansible/playbooks/50-configure-backups.yml |
| net-box_app_allow_first_deployment | net-box_app_first_deployment_confirmation | DEPLOY NETBOX ON VM 330                       | services/netbox/ansible/roles/netbox_app/tasks/main.yml    |

Source: [docs/reference/generated/gates.md](https://docs.reference/generated/gates.md)

## 21 Appendix D: Operator inputs

### 21.1 Operator inputs (L7)

Every secret or attended input the lab depends on: what it is, where it lives, and when it is used. **No value is ever stored in Git, printed, or handled by an agent.** Installed files are root-owned mode 0600 and checked by metadata only. OpenBao initialization and unsealing stay attended.

| Input                              | What it is                                                                                                    | Where it lives                                                                | Used at       |
|------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------|
| Proxmox root password hash         | crypt(3) hash for answer files                                                                                | Operator password manager; PVE_ROOT_PASSWORD_HASH at build time               | L0            |
| Root SSH public keys               | Keys baked into answer files                                                                                  | PVE_ROOT_SSH_KEYS_FILE (public keys only)                                     | L0            |
| Notification address               | mailto for answer files                                                                                       | PVE_MAILTO at build time                                                      | L0            |
| Node install inputs                | FQDN, disk and NIC filters                                                                                    | inventory/lab.yml nodes.*.install (not secret)                                | L0            |
| DSM admin credentials              | NAS administration                                                                                            | Operator password manager; never in automation                                | L0, L1        |
| Cluster SSH key                    | ~/.ssh/proxmox_cluster_ed25519                                                                                | Operator controller only                                                      | L1-L3, L8     |
| Root password prompt for pvecm add | Join authentication                                                                                           | Typed by the operator at the prompt                                           | L2            |
| Proxmox API token                  | Dedicated token for OpenTofu                                                                                  | TF_VAR_proxmox_api_token in the operator shell                                | L3            |
| State passphrase                   | Per-root PBKDF2/AES-GCM state and plan encryption                                                             | Operator password manager; TF_VAR_state_passphrase in local devcontainer only | L3, L5 (*arr) |
| State locations                    | D1 local backend and encrypted NAS copies                                                                     | D:\homelab-state\<root>\; NAS proxmox-cluster-backups/opentofu-state/<root>/  | L3            |
| Guest admin SSH keys               | Per-guest keys (homelab_services_ed25519, homelab_observability_ed25519, homelab_netbox_ed25519, OpenBao key) | Operator controller; public halves in rebuild.tfvars (git-ignored)            | L3-L5         |
| Guest host-key pins                | Fingerprints verified out of band (guest agent)                                                               | known_hosts; recorded in lab.yml                                              | L4            |
| VM 300 app secrets                 | NetKnife, Speedtest SMTP, DNS-01 credential, ...                                                              | /etc/homelab/secret-store/ on VM 300                                          | L5            |
| Observability secrets              | PVE exporter token, Grafana admin, SNMPv3, SMTP                                                               | /etc/observability/secret-store/ on VM 310                                    | L5            |
| NetBox superuser and API token     | NetBox administration                                                                                         | Operator-installed on VM 330                                                  | L5            |

| Input                       | What it is                                                                             | Where it lives                                                                                                                                  | Used at                                                            |
|-----------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| OpenBao TLS material        | Node certificates from the offline root                                                | Operator-installed on VMs 320-322                                                                                                               | L5                                                                 |
| Offline root CA key         | Signs OpenBao node certificates                                                        | <code>~/.local-credentials/home-lab/openbao/pki/</code> → move to encrypted custody                                                             | L5                                                                 |
| OpenBao unseal shares       | Five PGP-encrypted shares, threshold three                                             | D4: sole operator, five off-line keys/places; only public keys/fingerprints near Git or agents; never on NAS                                    | L7, attended 2026-10-10; verify VM 320-322 archives the day before |
| OpenBao backup age identity | Decrypts Raft snapshots                                                                | Outside OpenBao and the NAS                                                                                                                     | L6                                                                 |
| NAS CIFS credential (media) | <code>plex-reader/media</code> account for the guest mounts                            | <code>/etc/media/secrets/cifs.cred</code> on media-vm                                                                                           | L4                                                                 |
| Radarr / Prowlarr API keys  | App API keys                                                                           | <code>/etc/media/secrets/*.env</code> on the guest; <code>TF_VAR_radarr_api_key</code> , <code>TF_VAR_prowlarr_api_key</code> for the *arr root | L5                                                                 |
| qBittorrent Web UI login    | Download client credentials                                                            | <code>TF_VAR_qbittorrent_username</code> , <code>TF_VAR_qbittorrent_password</code> ; qBittorrent on Windows                                    | L5                                                                 |
| Plex claim                  | One-time claim in the operator's browser                                               | Never stored                                                                                                                                    | L5 (attended)                                                      |
| Plex token                  | <code>PLEX_TOKEN</code> for the library script                                         | Operator password manager; also inside <code>Preferences.xml</code> backups                                                                     | L5, L6                                                             |
| Media guest connection      | SSH forward port and admin user                                                        | <code>inventory/host_vars/media-vm.yml</code> (git-ignored)                                                                                     | L4-L6                                                              |
| Backup pull private key     | Dedicated passphrase-free ED25519 key; operator generates it                           | <code>~/.ssh/homelab_backup_pull_ed25519</code> on workstation only; agent handles only <code>.pub</code>                                       | L6                                                                 |
| Backup node public key      | Public ED25519 host key, compared with <code>lab.yml</code> fingerprint before pinning | <code>D:\homelab-backups\control\known_hosts</code> after verification                                                                          | L6                                                                 |
| Rotation list               | Credentials that must be rotated                                                       | <code>MEMORY.md</code> → Rotation list                                                                                                          | L7                                                                 |

Source: [docs/reference/operator-inputs.md](https://docs.reference/operator-inputs.md)

## 22 Appendix E: Versions and counts

### 22.1 Version matrix

| Kind            | Name                                    | Pinned                 | Source                                                                              |
|-----------------|-----------------------------------------|------------------------|-------------------------------------------------------------------------------------|
| Container image | corentinh/it-tools:2024.10.22-7ca5933   | sha256:6f177c156b94... | services/homelab-services/config/templates/stage-d/compose/it-tools/compose.yaml    |
| Container image | ghcr.io/gethomepage/homepage:v1.13.2    | sha256:c881120b024d... | services/homelab-services/config/templates/stage-d/compose/homepage/compose.yaml    |
| Container image | ghcr.io/google/cadvisor:v0.60.5         | (no digest)            | observability/docker/compose.yaml                                                   |
| Container image | ghcr.io/onedr0p/expo-tarr:v2.3.0        | sha256:af535d94061c... | services/media/config/compose.yaml                                                  |
| Container image | ghcr.io/onedr0p/expo-tarr:v2.3.0        | sha256:af535d94061c... | services/media/config/compose.yaml                                                  |
| Container image | ghcr.io/recyclarr/recyclarr:8           | sha256:6e69e009e1cd... | services/media/config/compose.yaml                                                  |
| Container image | ghcr.io/schlagmichdoch/pairdrop:v1.11.2 | sha256:39fdf4aa90c5... | services/homelab-services/config/templates/stage-d/compose/pairdrop/compose.yaml    |
| Container image | grafana/alloy:v1.18.0                   | (no digest)            | observability/docker/compose.yaml                                                   |
| Container image | grafana/alloy:v1.18.0                   | sha256:491b0578c049... | services/media/config/compose.yaml                                                  |
| Container image | grafana/grafana:13.1.0                  | (no digest)            | observability/docker/compose.yaml                                                   |
| Container image | grafana/loki:3.7.4                      | (no digest)            | observability/docker/compose.yaml                                                   |
| Container image | homelab/netknife-api:28ef127baeab       | (no digest)            | services/homelab-services/config/templates/stage-d/compose/netknife/compose.yaml    |
| Container image | homelab/netknife-web:28ef127baeab       | (no digest)            | services/homelab-services/config/templates/stage-d/compose/netknife/compose.yaml    |
| Container image | jc21/nginx-proxy-manager:2.15.1         | sha256:99a885f56ca2... | services/homelab-services/config/templates/stage-d/compose/npm/compose.yaml         |
| Container image | louislam/uptime-kuma:2.3.2              | sha256:dbede95c5296... | services/homelab-services/config/templates/stage-d/compose/uptime-kuma/compose.yaml |

| Kind              | Name                                         | Pinned                 | Source                                                                                    |
|-------------------|----------------------------------------------|------------------------|-------------------------------------------------------------------------------------------|
| Container image   | lscr.io/linuxserver/prowlarr:latest          | sha256:f2b26429893d... | services/media/config/compose.yaml                                                        |
| Container image   | lscr.io/linuxserver/radarr:latest            | sha256:adb6c09d6b72... | services/media/config/compose.yaml                                                        |
| Container image   | lscr.io/linuxserver/speedtest-tracker:1.14.3 | sha256:cdb782713a9c... | services/homelab-services/config/templates/stage-d/compose/speedtest-tracker/compose.yaml |
| Container image   | prom/alertmanager:v0.32.1                    | (no digest)            | observability/docker/compose.yaml                                                         |
| Container image   | prom/blackbox-exporter:v0.28.0               | (no digest)            | observability/docker/compose.yaml                                                         |
| Container image   | prom/node-exporter:v1.12.1                   | (no digest)            | observability/docker/compose.yaml                                                         |
| Container image   | prom/node-exporter:v1.12.1                   | sha256:1b4e4438faca... | services/media/config/compose.yaml                                                        |
| Container image   | prom/prometheus:v3.12.0                      | (no digest)            | observability/docker/compose.yaml                                                         |
| Container image   | prom/snmp-exporter:v0.30.1                   | (no digest)            | observability/docker/compose.yaml                                                         |
| Container image   | prompve/prometheus-pve-exporter:3.9.0        | (no digest)            | observability/docker/compose.yaml                                                         |
| Container image   | sissbruecker/linkding:1.45.0                 | sha256:5bc84e076626... | services/homelab-services/config/templates/stage-d/compose/linkding/compose.yaml          |
| Galaxy collection | ansible.posix                                | >=1.5.4                | platform/proxmox/requirements.yml                                                         |
| Galaxy collection | ansible.posix                                | 2.2.2                  | services/media/requirements.yml                                                           |
| Galaxy collection | ansible.utils                                | >=5.1.0                | platform/proxmox/requirements.yml                                                         |
| Galaxy collection | ansible.windows                              | 3.8.0                  | services/media/requirements.yml                                                           |
| Galaxy collection | community.docker                             | 5.0.6                  | observability/requirements.yml                                                            |
| Galaxy collection | community.docker                             | 5.0.6                  | services/media/requirements.yml                                                           |
| Galaxy collection | community.general                            | 12.4.0                 | observability/requirements.yml                                                            |
| Galaxy collection | community.general                            | >=9.0.0                | platform/proxmox/requirements.yml                                                         |
| Galaxy collection | community.general                            | 12.4.0                 | services/media/requirements.yml                                                           |
| OpenTofu provider | bpg/proxmox                                  | 0.107.0                | platform/proxmox/terraform/homelab-guests/versions.tf                                     |

| Kind              | Name               | Pinned    | Source                                                |
|-------------------|--------------------|-----------|-------------------------------------------------------|
| OpenTofu provider | bpg/proxmox        | 0.107.0   | platform/proxmox/terraform/rebuild-guests/versions.tf |
| OpenTofu provider | bpg/proxmox        | 0.107.0   | security/secrets-openbao/terraform/versions.tf        |
| OpenTofu provider | bpg/proxmox        | 0.107.0   | services/netbox/terraform/proxmox-guest/versions.tf   |
| OpenTofu provider | devopsarr/prowlarr | 3.2.1     | services/media/terraform/arr/versions.tf              |
| OpenTofu provider | devopsarr/radarr   | 2.5.0     | services/media/terraform/arr/versions.tf              |
| Python package    | ansible-core       | ==2.19.4  | observability/requirements.txt                        |
| Python package    | ansible-core       | >=2.16    | platform/proxmox/requirements.txt                     |
| Python package    | ansible-core       | ==2.19.2  | security/secrets-openbao/requirements.txt             |
| Python package    | ansible-core       | ==2.19.4  | services/media/requirements.txt                       |
| Python package    | ansible-lint       | ==25.12.2 | observability/requirements.txt                        |
| Python package    | ansible-lint       | >=24.2.0  | platform/proxmox/requirements.txt                     |
| Python package    | ansible-lint       | ==25.8.2  | security/secrets-openbao/requirements.txt             |
| Python package    | ansible-lint       | ==25.12.2 | services/media/requirements.txt                       |
| Python package    | jmespath           | >=1.0.1   | platform/proxmox/requirements.txt                     |
| Python package    | netaddr            | >=1.2.1   | platform/proxmox/requirements.txt                     |
| Python package    | plexapi            | ==4.18.3  | services/media/requirements.txt                       |
| Python package    | pre-commit         | ==4.3.0   | observability/requirements.txt                        |
| Python package    | pytest             | >=8.0.0   | platform/proxmox/requirements.txt                     |
| Python package    | pytest             | ==8.4.2   | services/media/requirements.txt                       |
| Python package    | pytest             | ==8.4.2   | tests/requirements.txt                                |
| Python package    | PyYAML             | ==6.0.3   | services/media/requirements.txt                       |
| Python package    | PyYAML             | ==6.0.3   | tests/requirements.txt                                |
| Python package    | ruff               | >=0.5.0   | platform/proxmox/requirements.txt                     |

| Kind           | Name         | Pinned    | Source                                    |
|----------------|--------------|-----------|-------------------------------------------|
| Python package | ruff         | ==0.12.11 | security/secrets-openbao/requirements.txt |
| Python package | ruff         | ==0.12.11 | services/media/requirements.txt           |
| Python package | ruff         | ==0.12.11 | tests/requirements.txt                    |
| Python package | yamllint     | ==1.37.1  | observability/requirements.txt            |
| Python package | yamllint     | >=1.35.1  | platform/proxmox/requirements.txt         |
| Python package | yamllint     | ==1.37.1  | security/secrets-openbao/requirements.txt |
| Python package | yamllint     | ==1.37.1  | services/media/requirements.txt           |
| devcontainer   | ansible-core | 2.19.4    | .devcontainer/Dockerfile                  |
| devcontainer   | ansible-lint | 25.12.2   | .devcontainer/Dockerfile                  |
| devcontainer   | gitleaks     | 8.30.1    | .devcontainer/Dockerfile                  |
| devcontainer   | mermaid-cli  | 12.0.0    | .devcontainer/Dockerfile                  |
| devcontainer   | node         | 24.21.0   | .devcontainer/Dockerfile                  |
| devcontainer   | opentofu     | 1.12.6    | .devcontainer/Dockerfile                  |
| devcontainer   | pandoc       | 3.12      | .devcontainer/Dockerfile                  |
| devcontainer   | pre-commit   | 4.3.0     | .devcontainer/Dockerfile                  |
| devcontainer   | pytest       | 8.4.2     | .devcontainer/Dockerfile                  |
| devcontainer   | ruff         | 0.12.11   | .devcontainer/Dockerfile                  |
| devcontainer   | weasyprint   | 70.0      | .devcontainer/Dockerfile                  |
| devcontainer   | yamllint     | 1.37.1    | .devcontainer/Dockerfile                  |

Source: <docs/reference/generated/versions.md>

## 22.2 Repository counts

| Item                                           | Count |
|------------------------------------------------|-------|
| Components with a Makefile                     | 9     |
| Ansible playbooks                              | 56    |
| Ansible roles                                  | 42    |
| OpenTofu roots                                 | 6     |
| Python test functions (before parametrisation) | 289   |
| OpenTofu test runs (mocked providers)          | 5     |
| GitHub Actions workflows                       | 2     |
| Architecture decision records                  | 9     |
| Agent skills                                   | 10    |

| Item                            | Count |
|---------------------------------|-------|
| Walkthrough diagrams            | 17    |
| Proxmox nodes (lab.yml)         | 3     |
| Guests (lab.yml)                | 8     |
| Scheduled vzdump jobs (lab.yml) | 7     |

Source: <docs/reference/generated/counts.md>